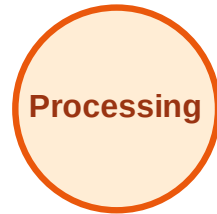


BFS Traversal

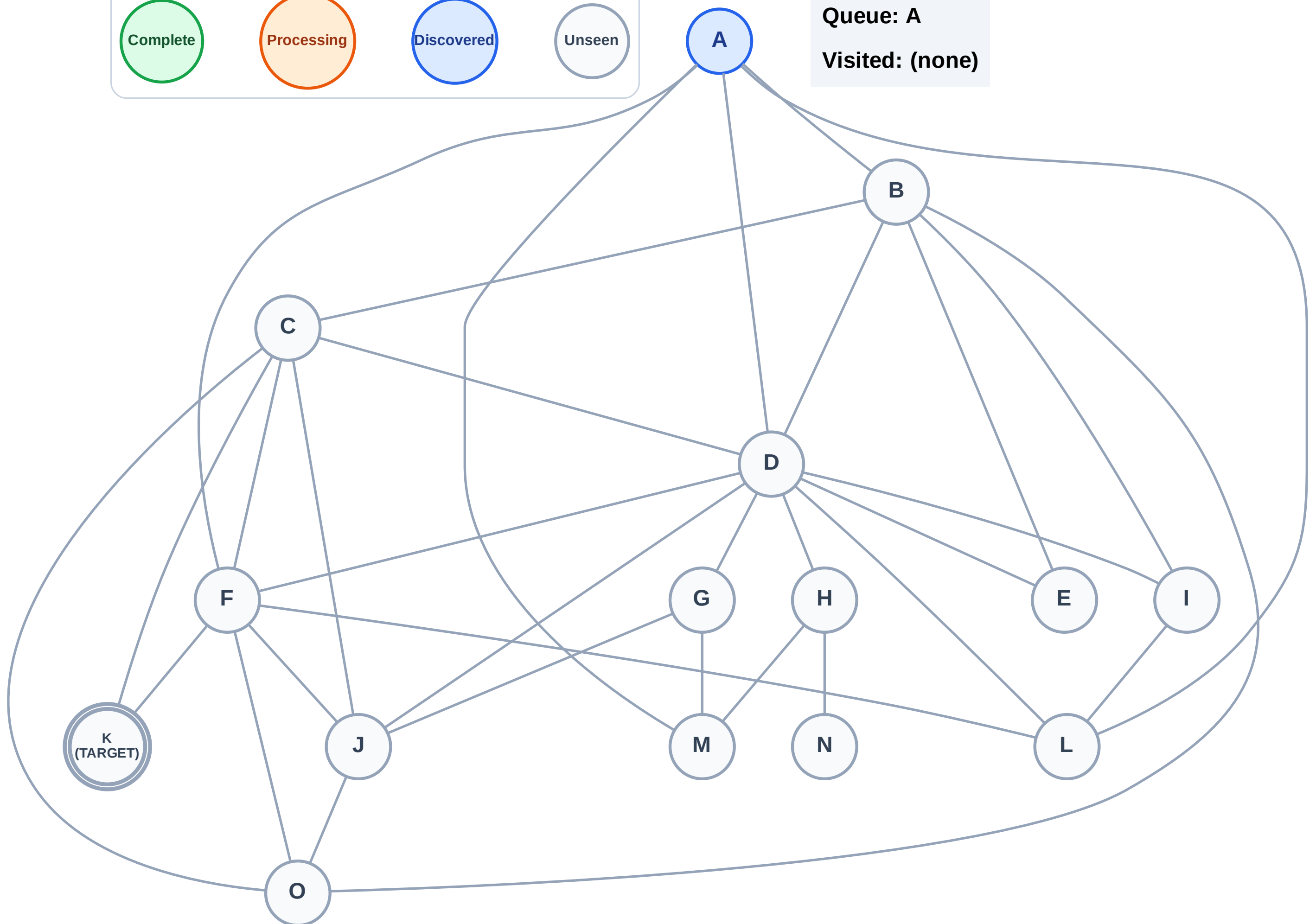
Step 1: Start at A (Seeking target node K)

Legend



Queue: A

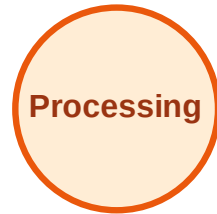
Visited: (none)



BFS Traversal

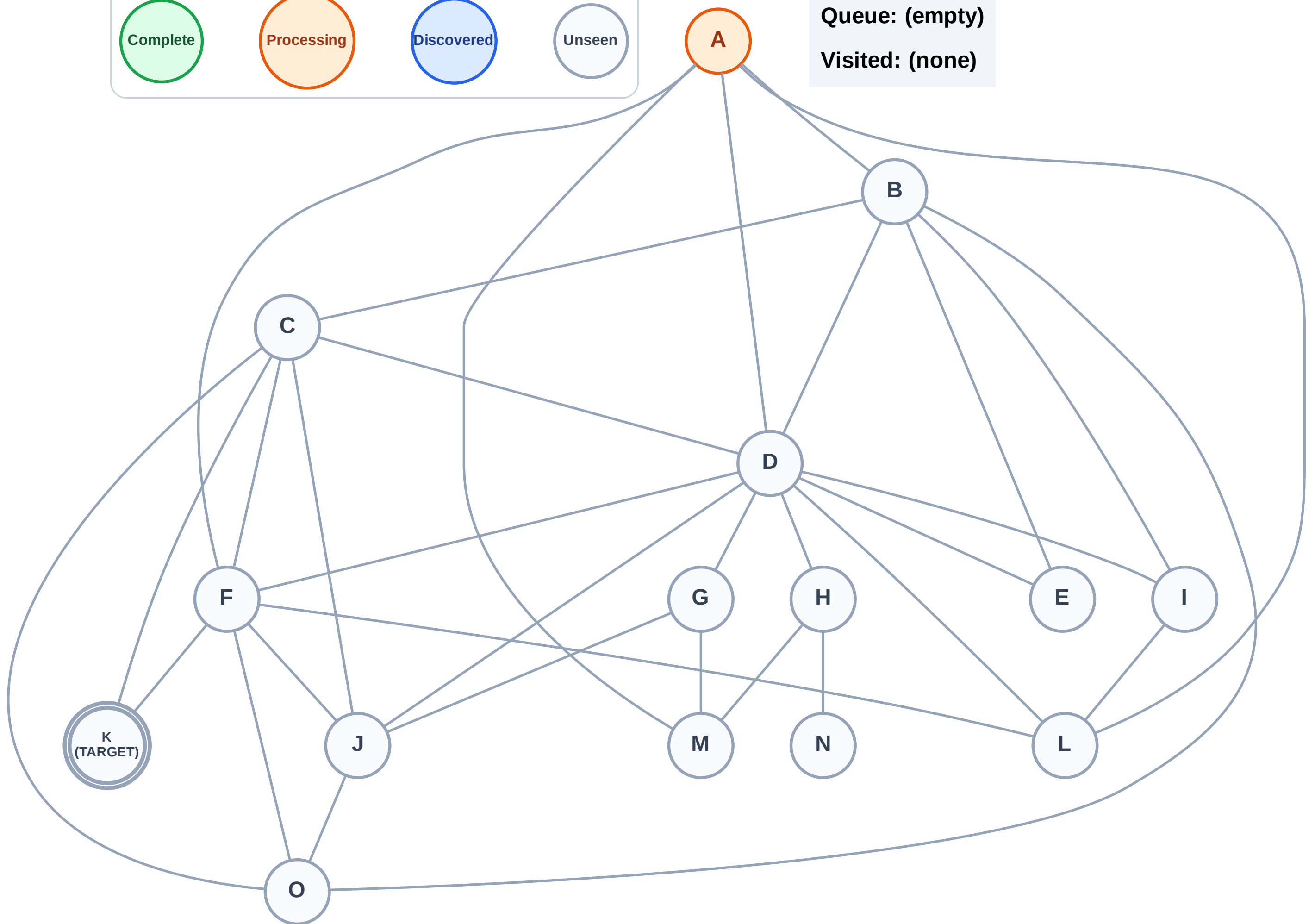
Step 2: Process node A

Legend



Queue: (empty)

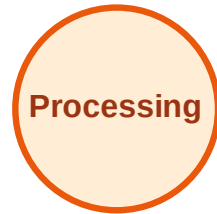
Visited: (none)



BFS Traversal

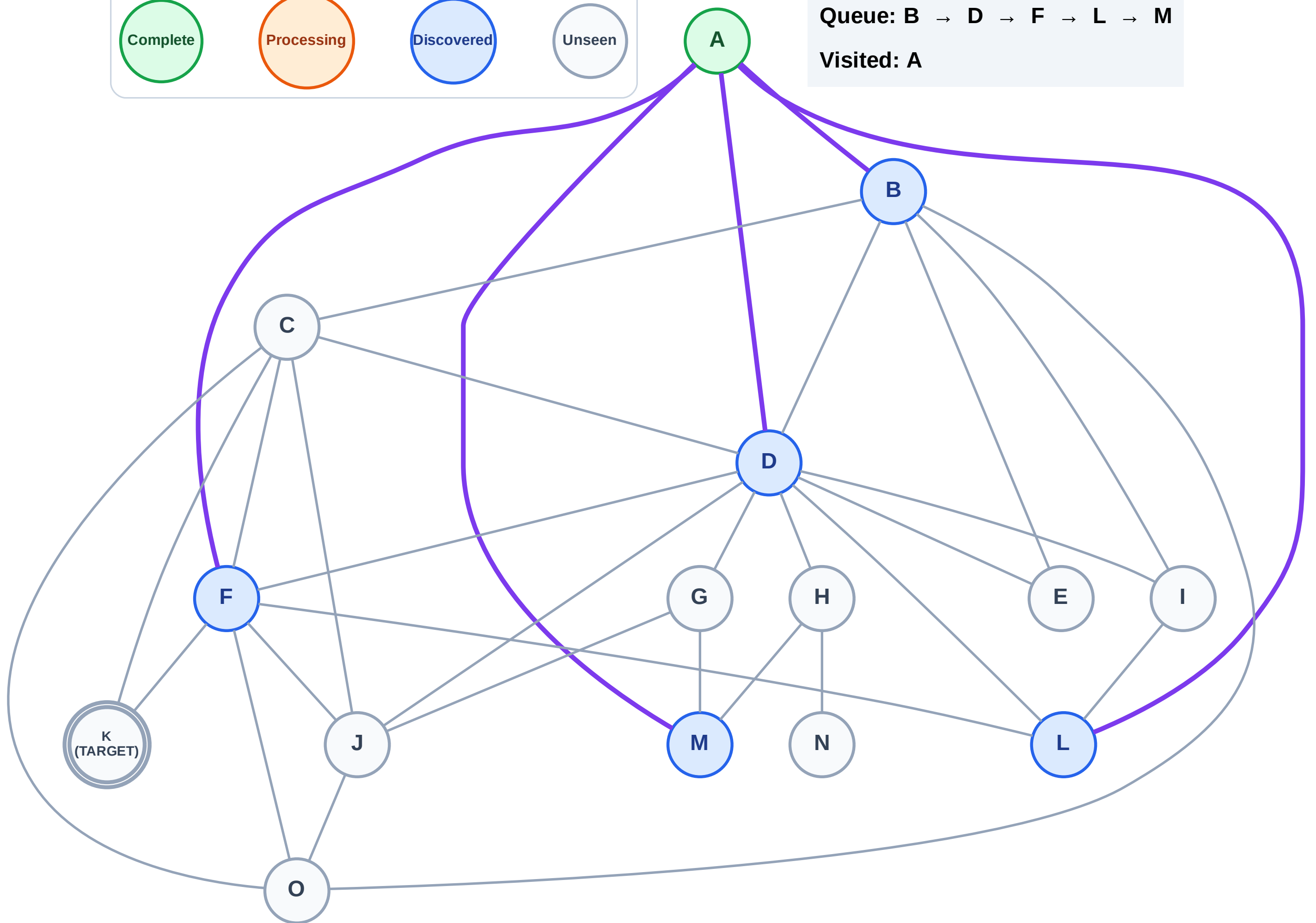
Step 3: Discover B, D, F, L, M from A

Legend



Queue: B → D → F → L → M

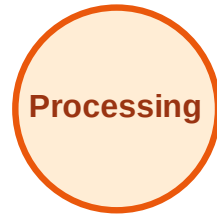
Visited: A



BFS Traversal

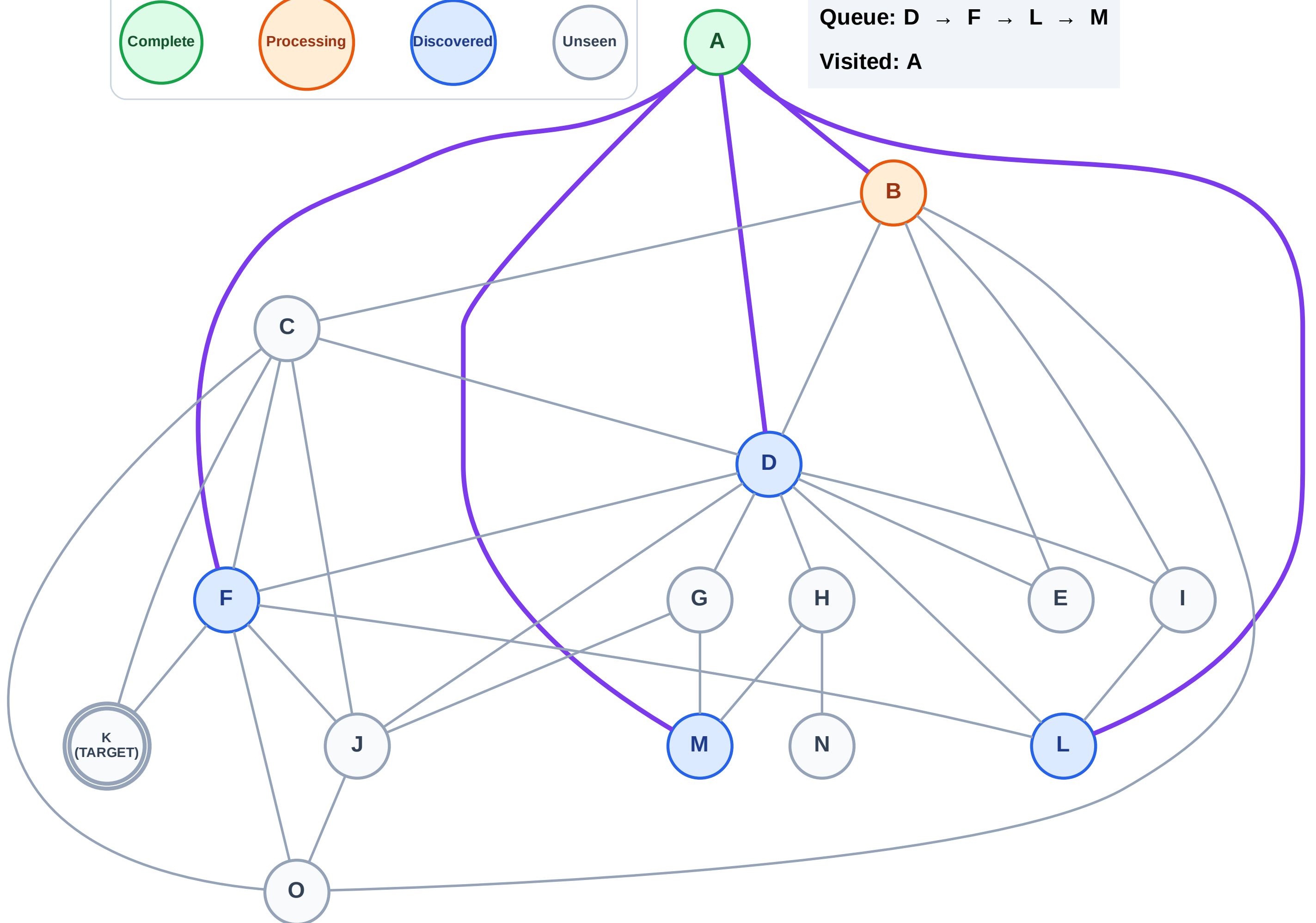
Step 4: Process node B

Legend



Queue: D → F → L → M

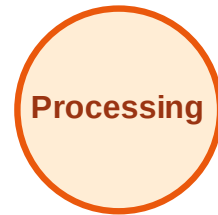
Visited: A



BFS Traversal

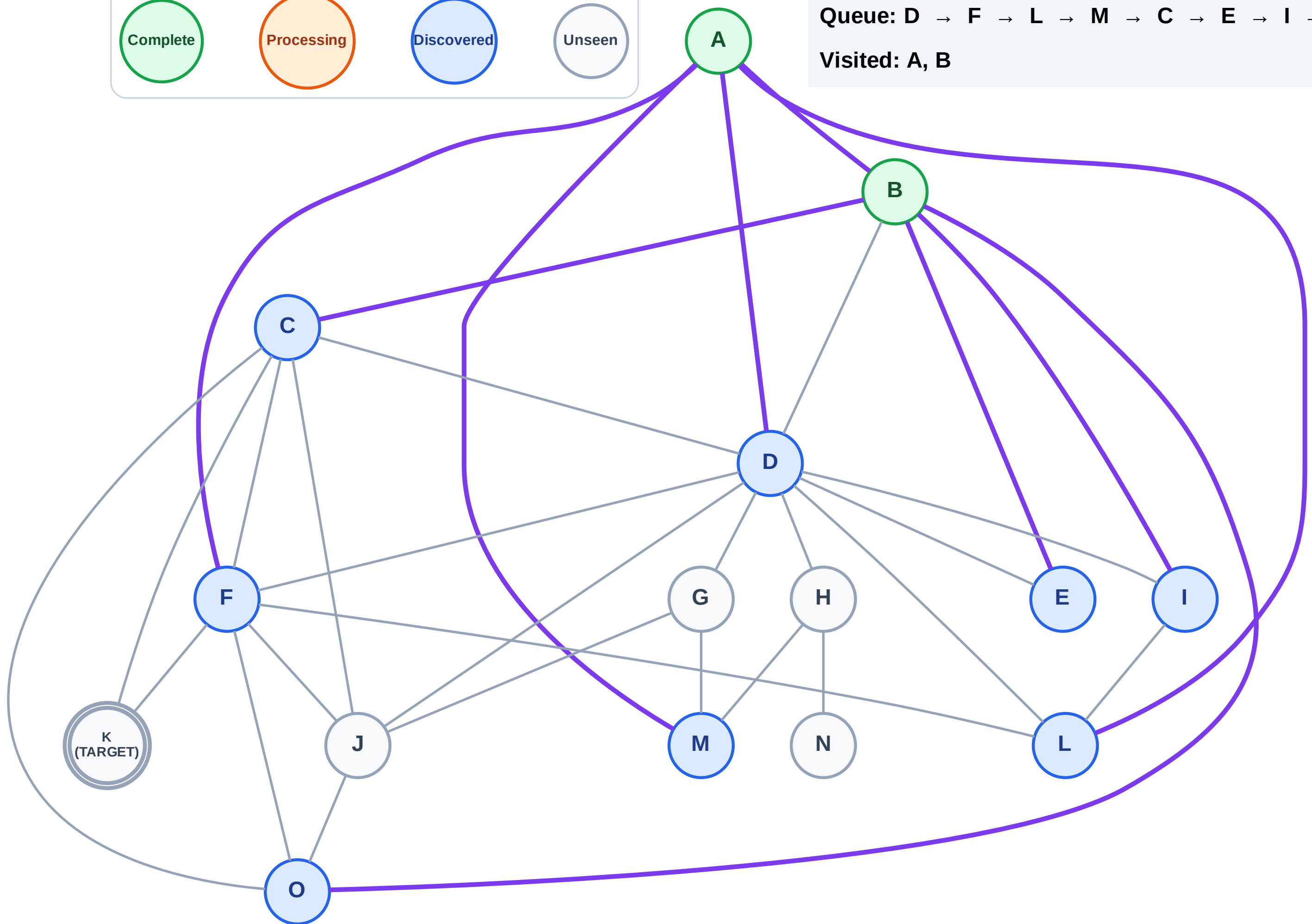
Step 5: Discover C, E, I, O from B

Legend



Queue: D → F → L → M → C → E → I → O

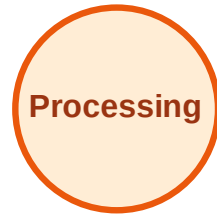
Visited: A, B



BFS Traversal

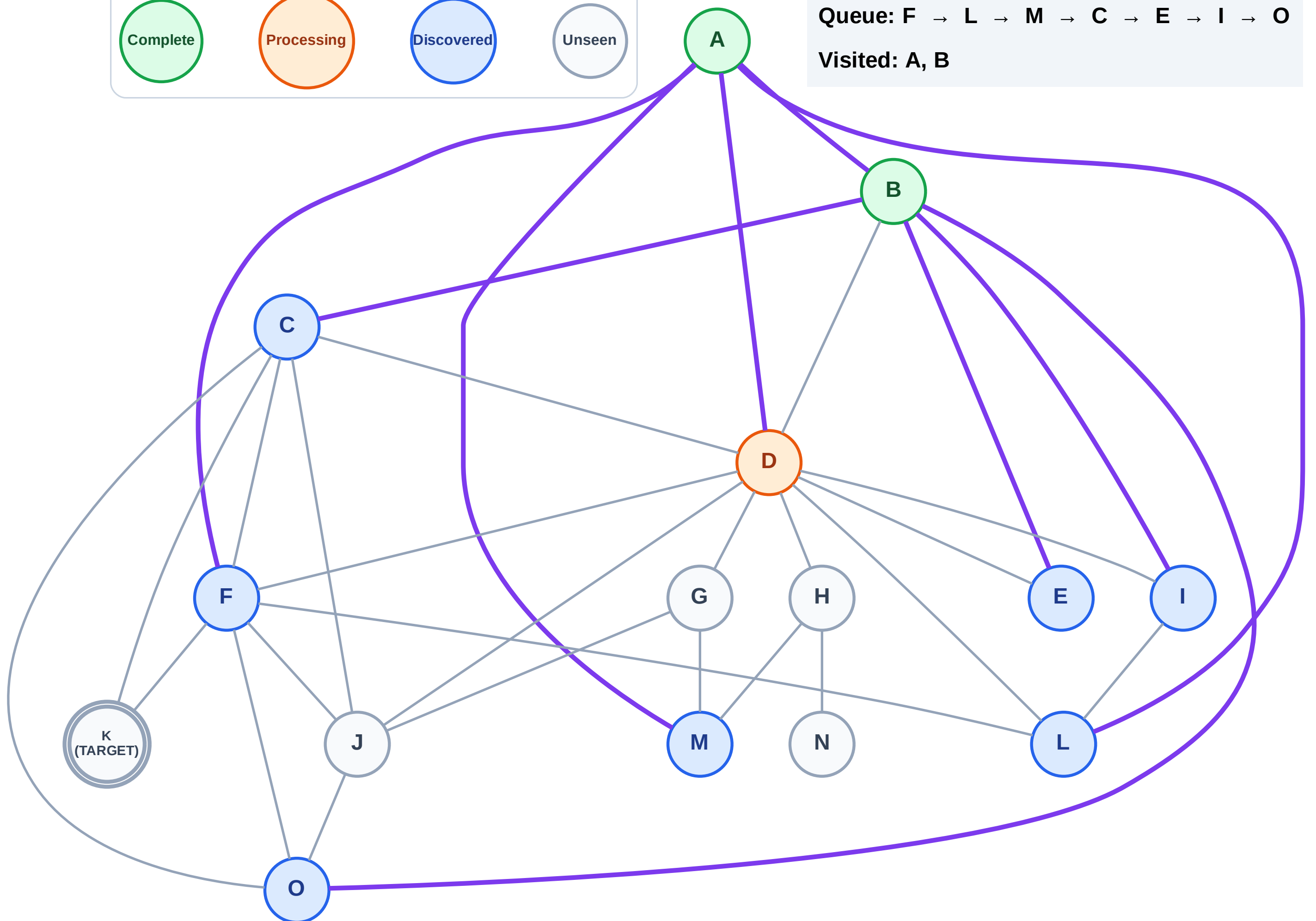
Step 6: Process node D

Legend



Queue: F → L → M → C → E → I → O

Visited: A, B



BFS Traversal

Step 7: Discover G, H, J from D

Legend

Complete

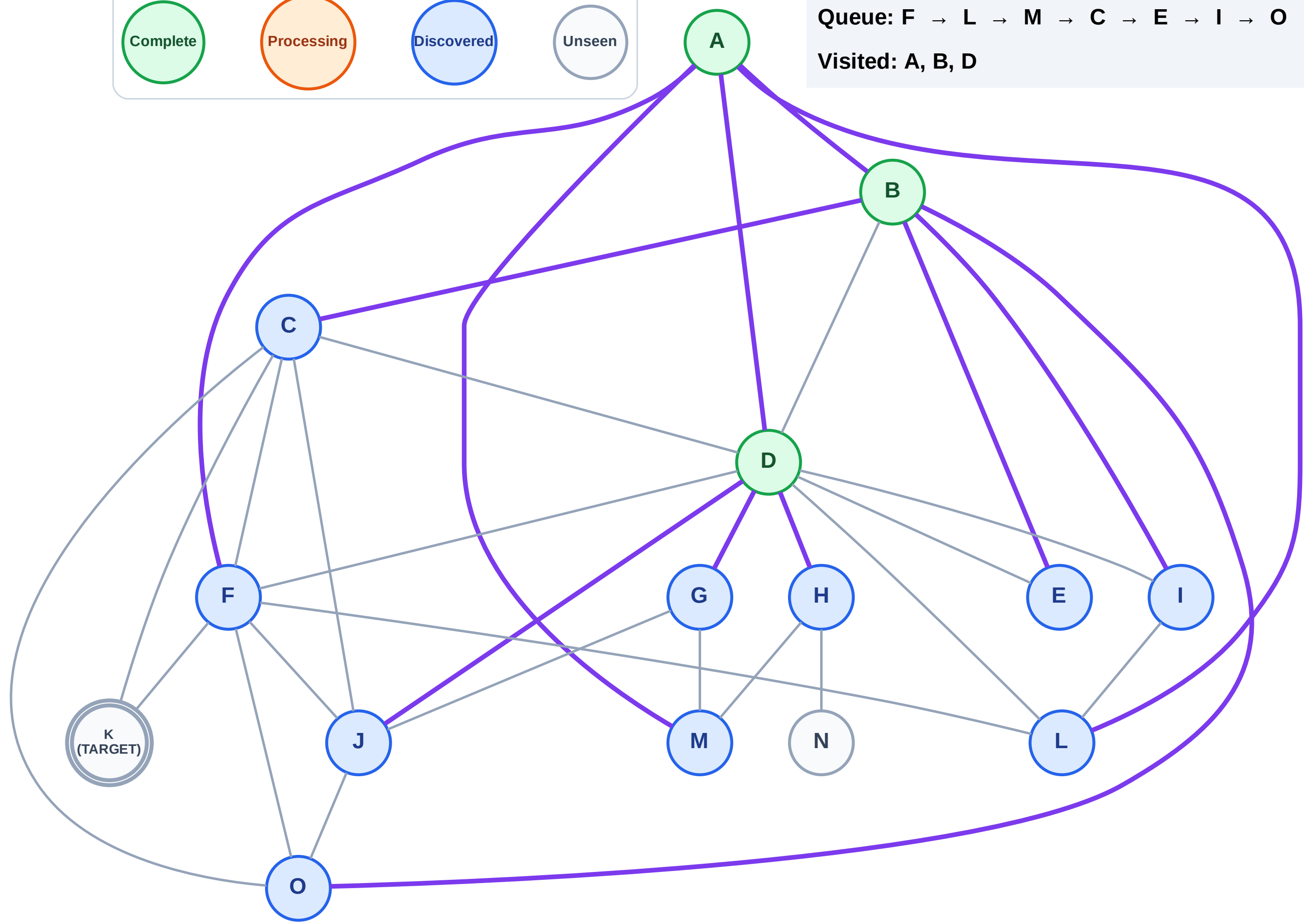
Processing

Discovered

Unseen

Queue: F → L → M → C → E → I → O → G → H → J

Visited: A, B, D



BFS Traversal

Step 8: Process node F

Legend

Complete

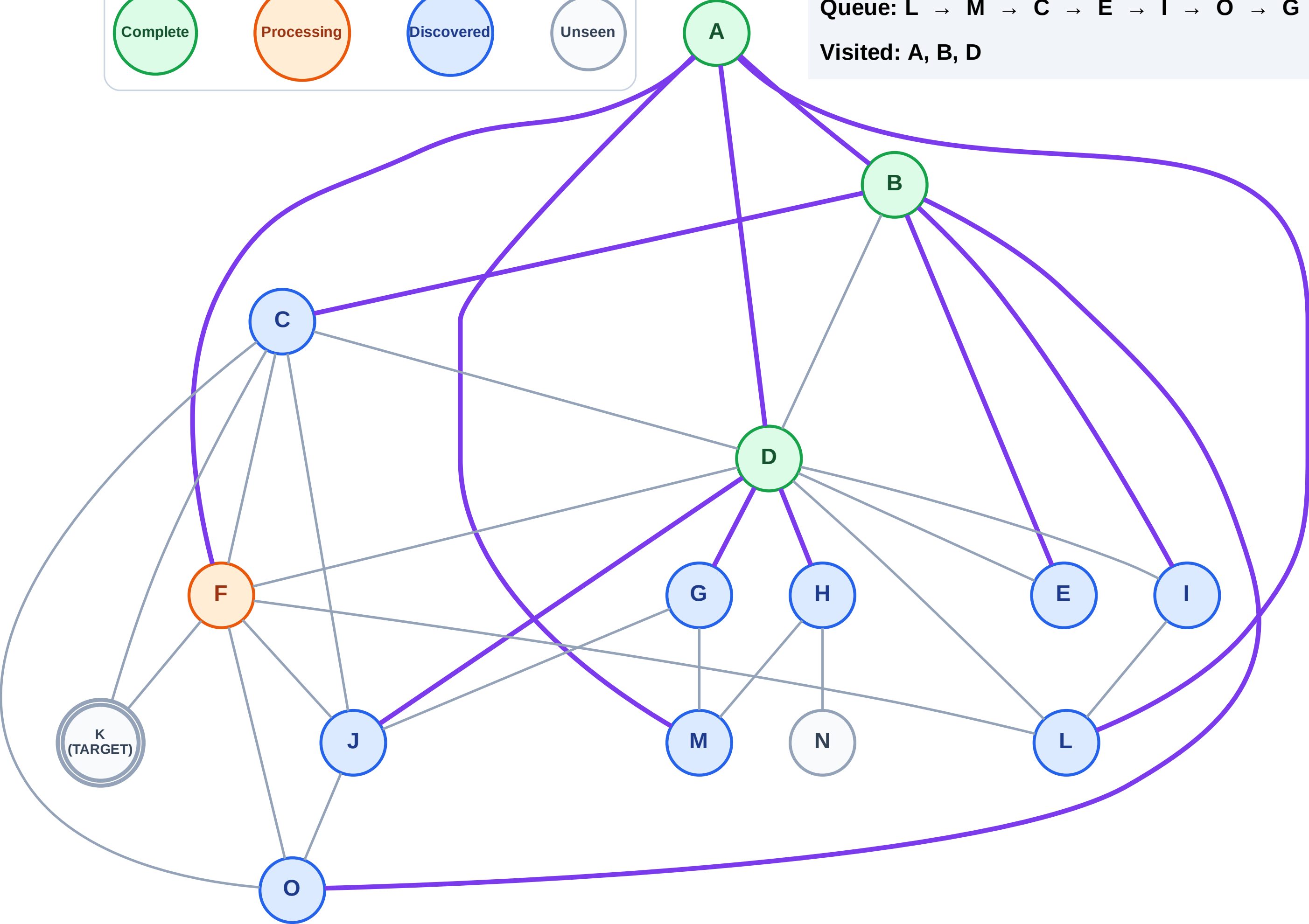
Processing

Discovered

Unseen

Queue: L → M → C → E → I → O → G → H → J

Visited: A, B, D



BFS Traversal

Step 9: Discover K from F

Legend

Complete

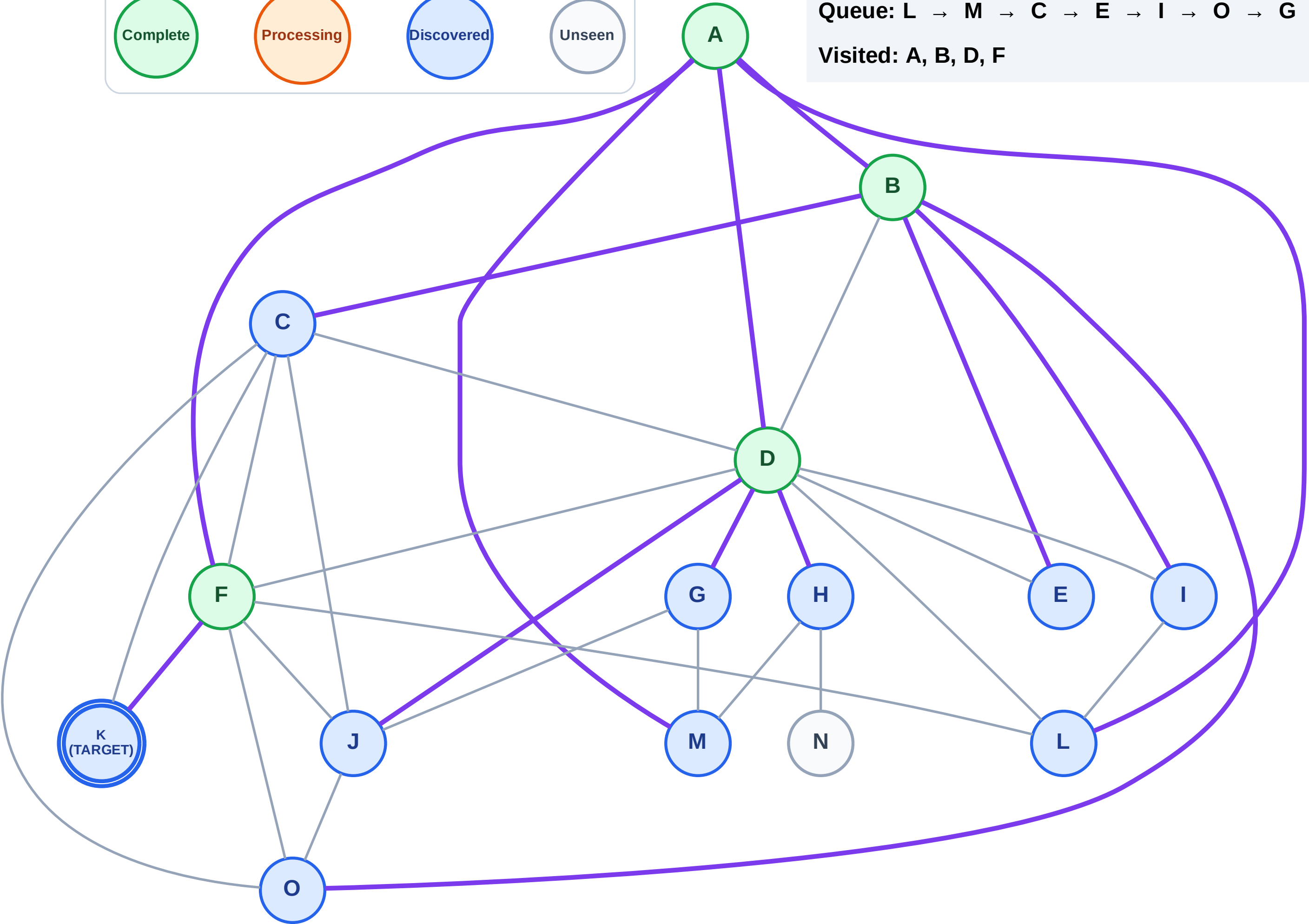
Processing

Discovered

Unseen

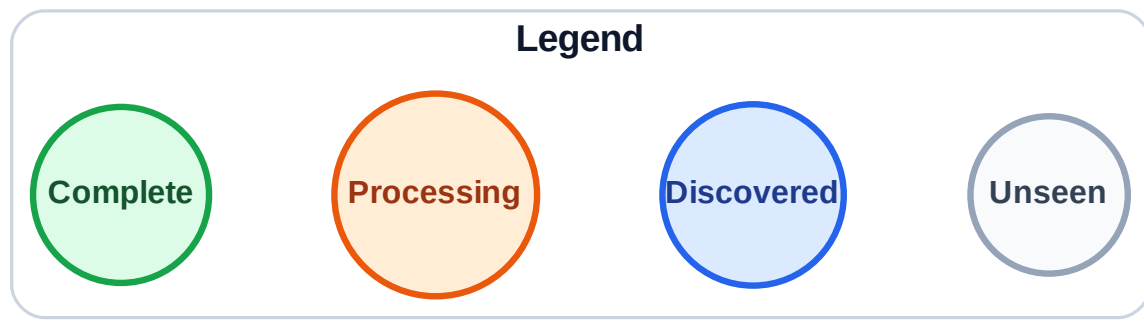
Queue: L → M → C → E → I → O → G → H → J → K

Visited: A, B, D, F

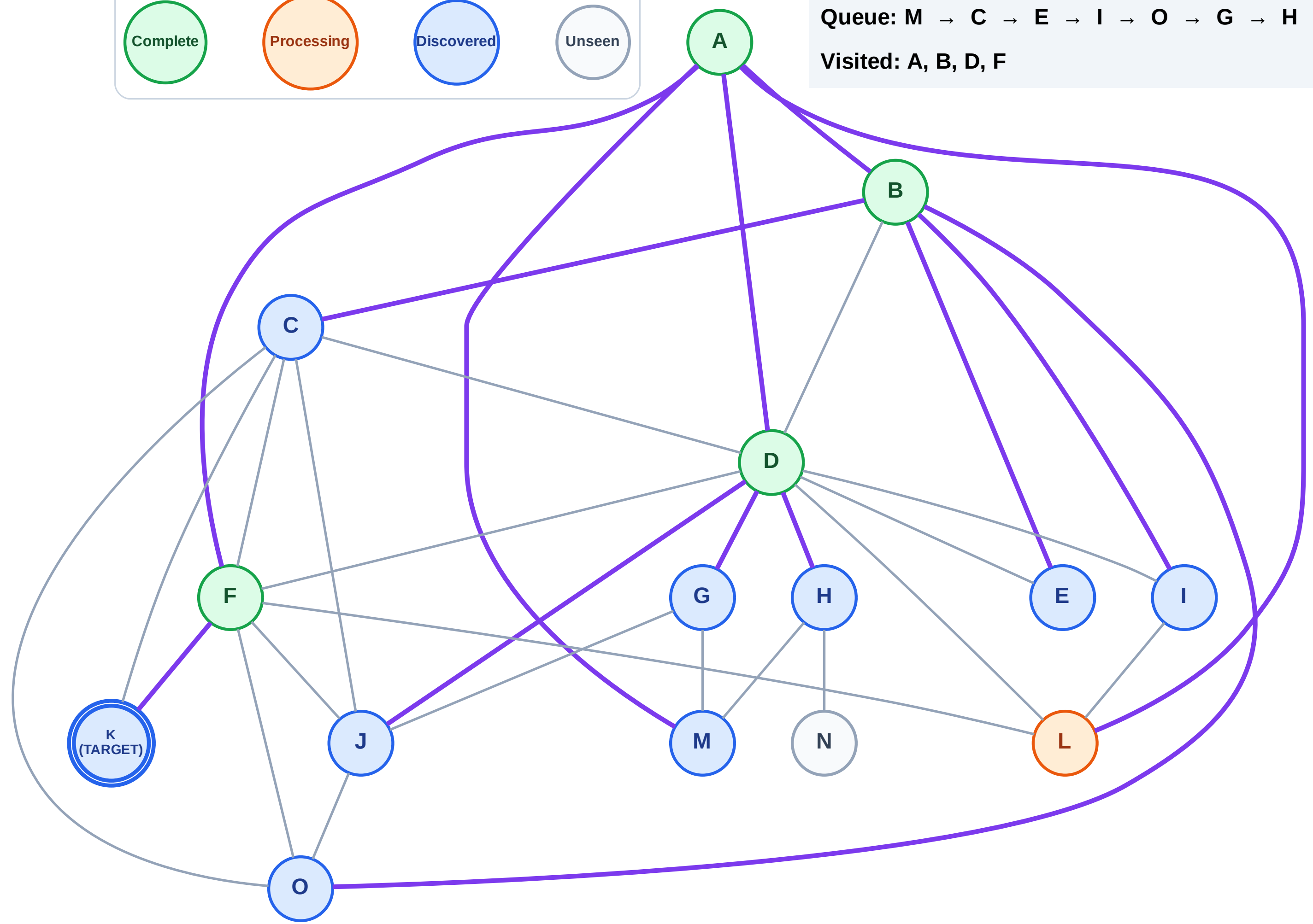


BFS Traversal

Step 10: Process node L

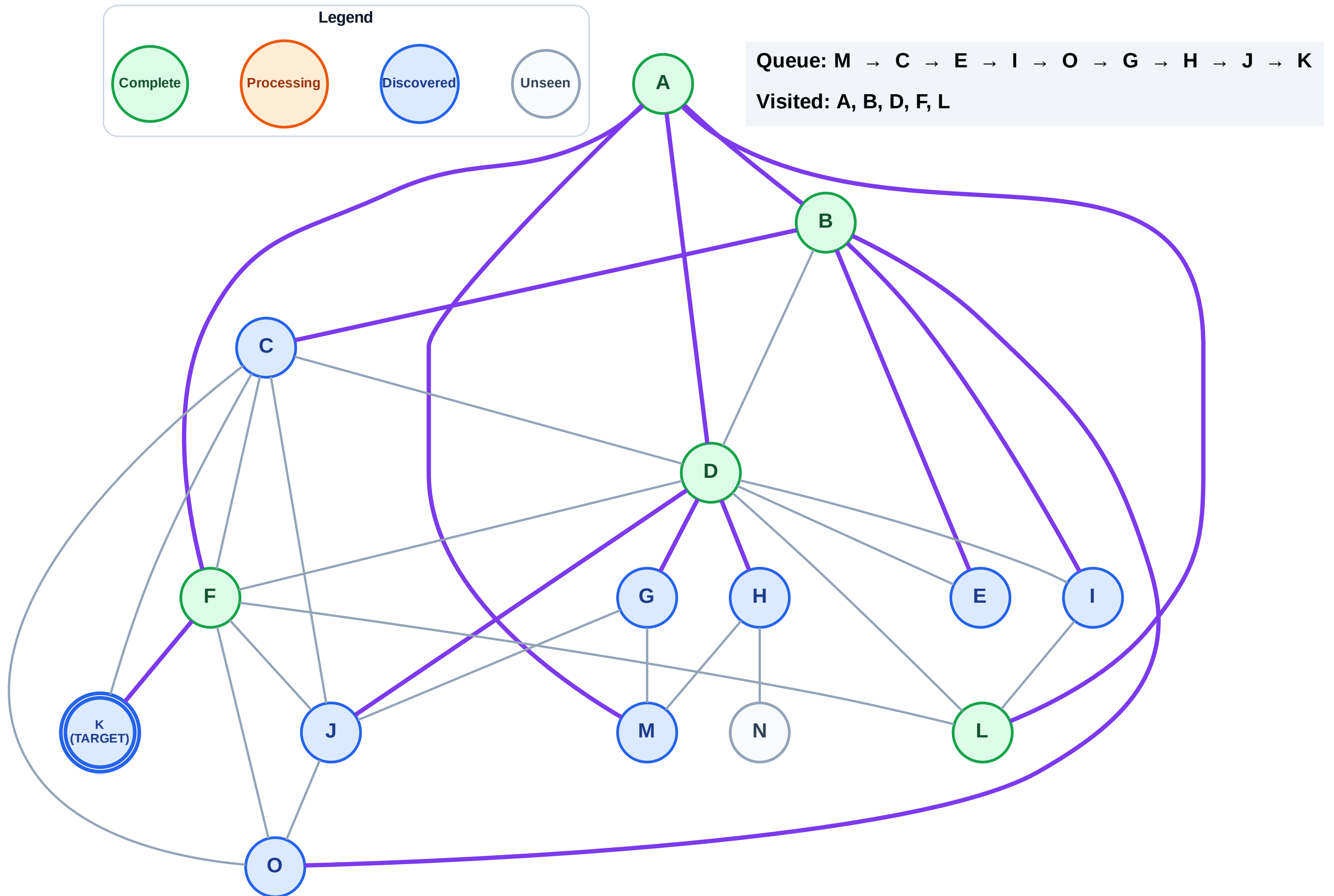


Queue: M → C → E → I → O → G → H → J → K
Visited: A, B, D, F



Step 11: No new neighbors from L

Step 11: No new neighbors from L



BFS Traversal

Step 12: Process node M

Legend

Complete

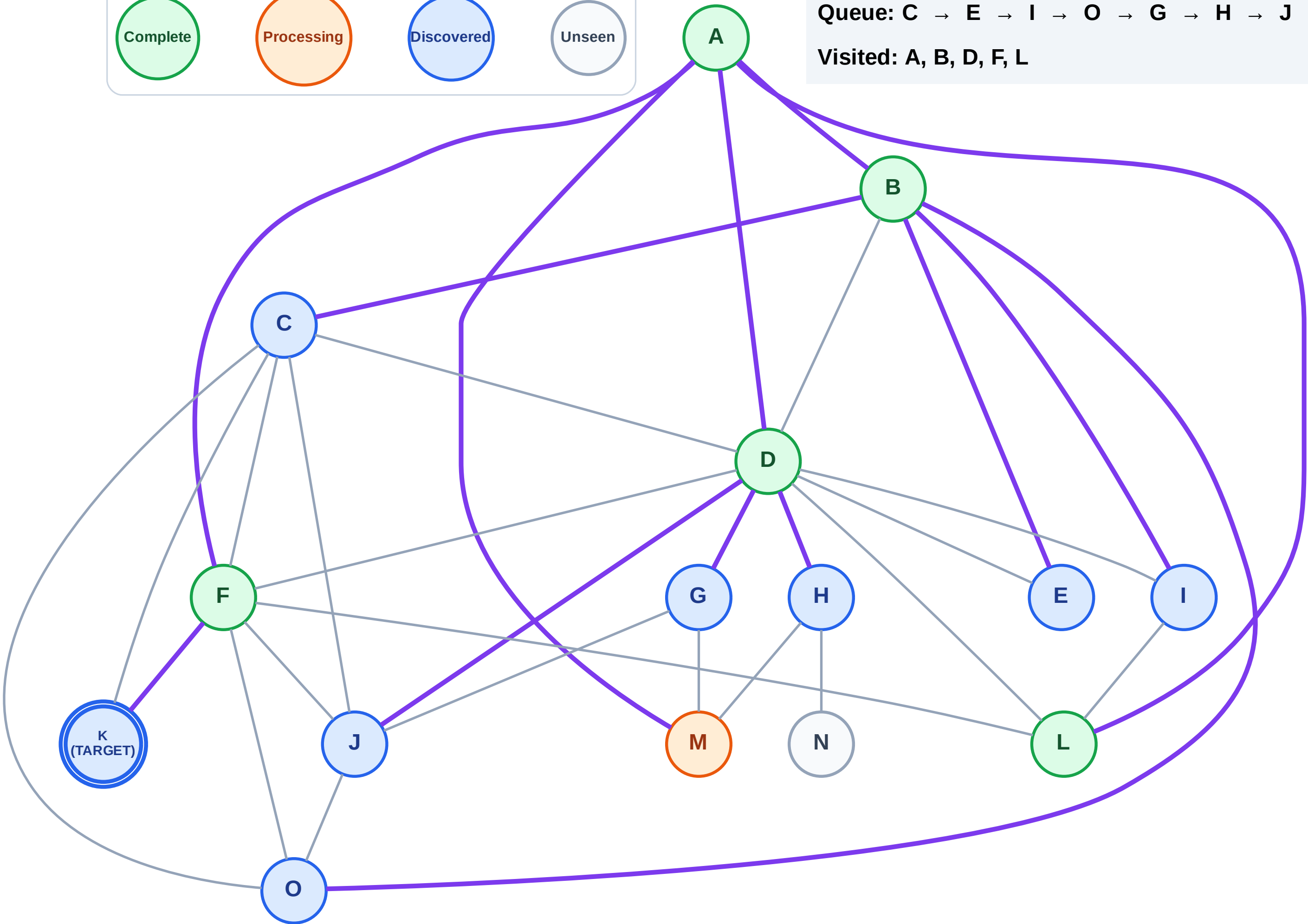
Processing

Discovered

Unseen

Queue: C → E → I → O → G → H → J → K

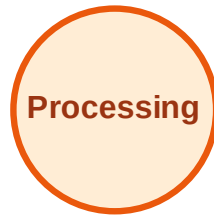
Visited: A, B, D, F, L



BFS Traversal

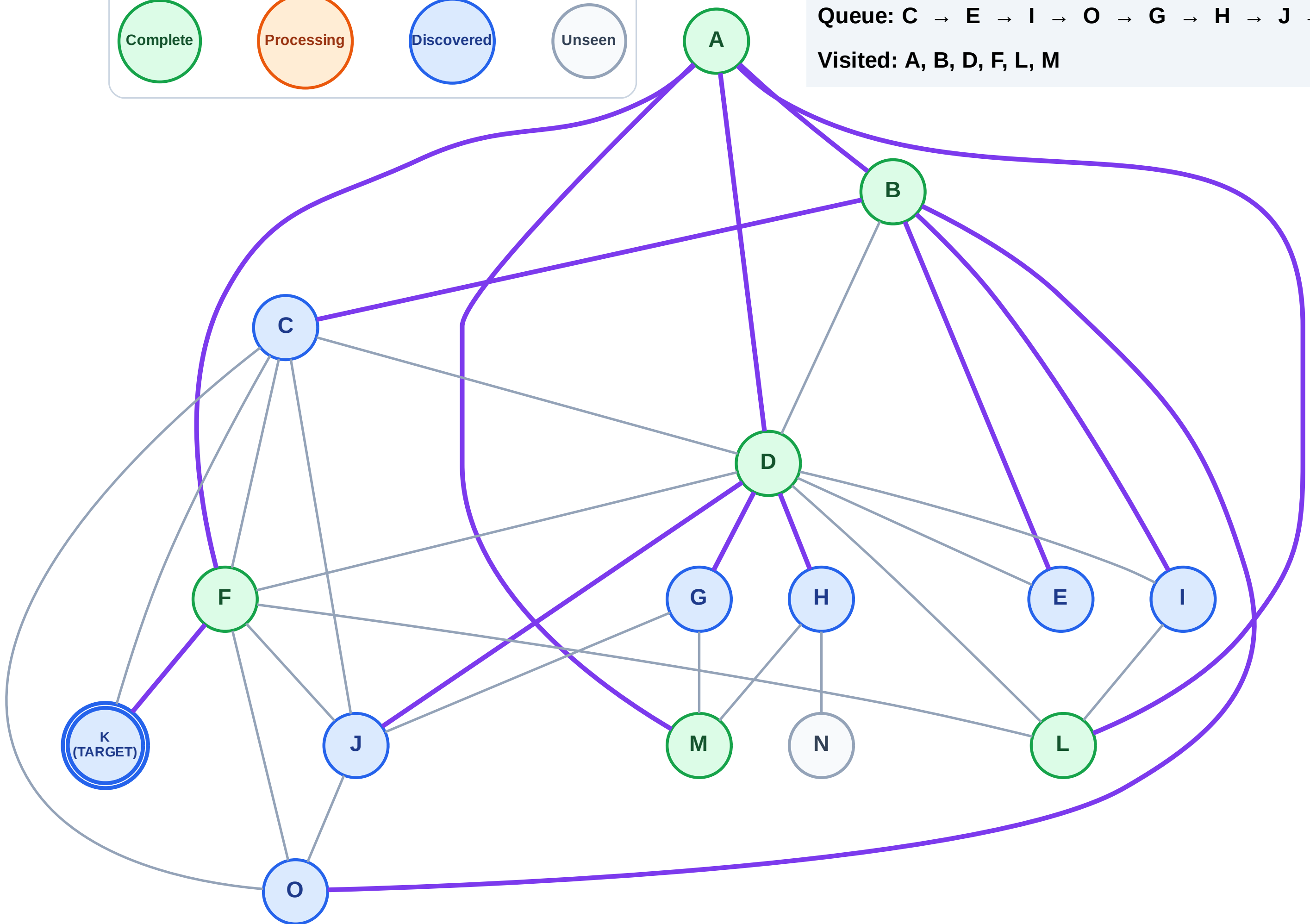
Step 13: No new neighbors from M

Legend



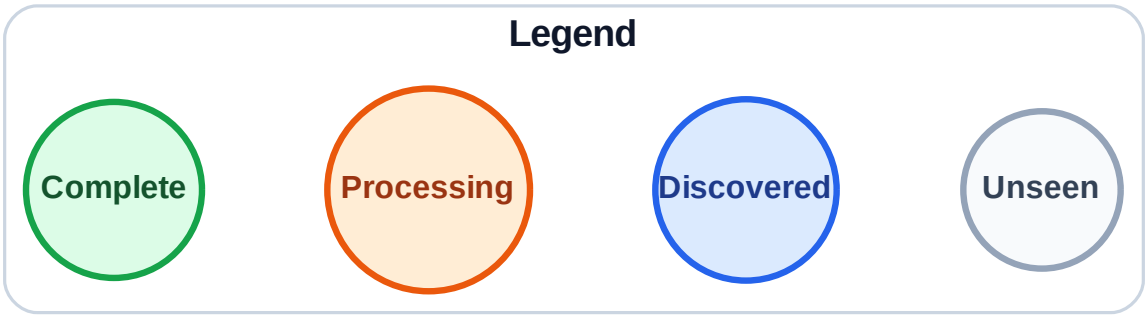
Queue: C → E → I → O → G → H → J → K

Visited: A, B, D, F, L, M

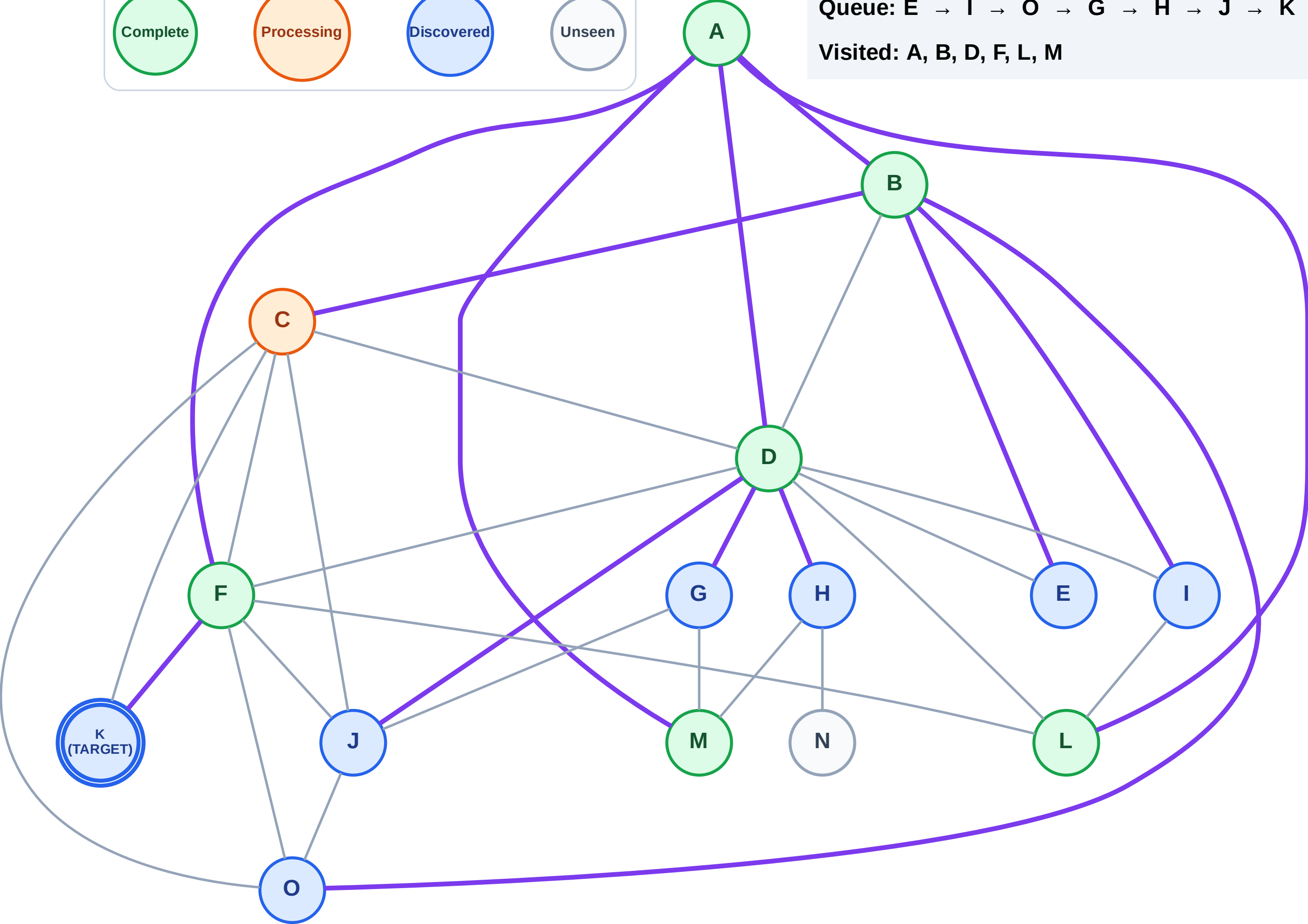


BFS Traversal

Step 14: Process node C



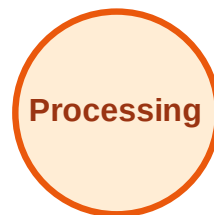
Queue: E → I → O → G → H → J → K
Visited: A, B, D, F, L, M



BFS Traversal

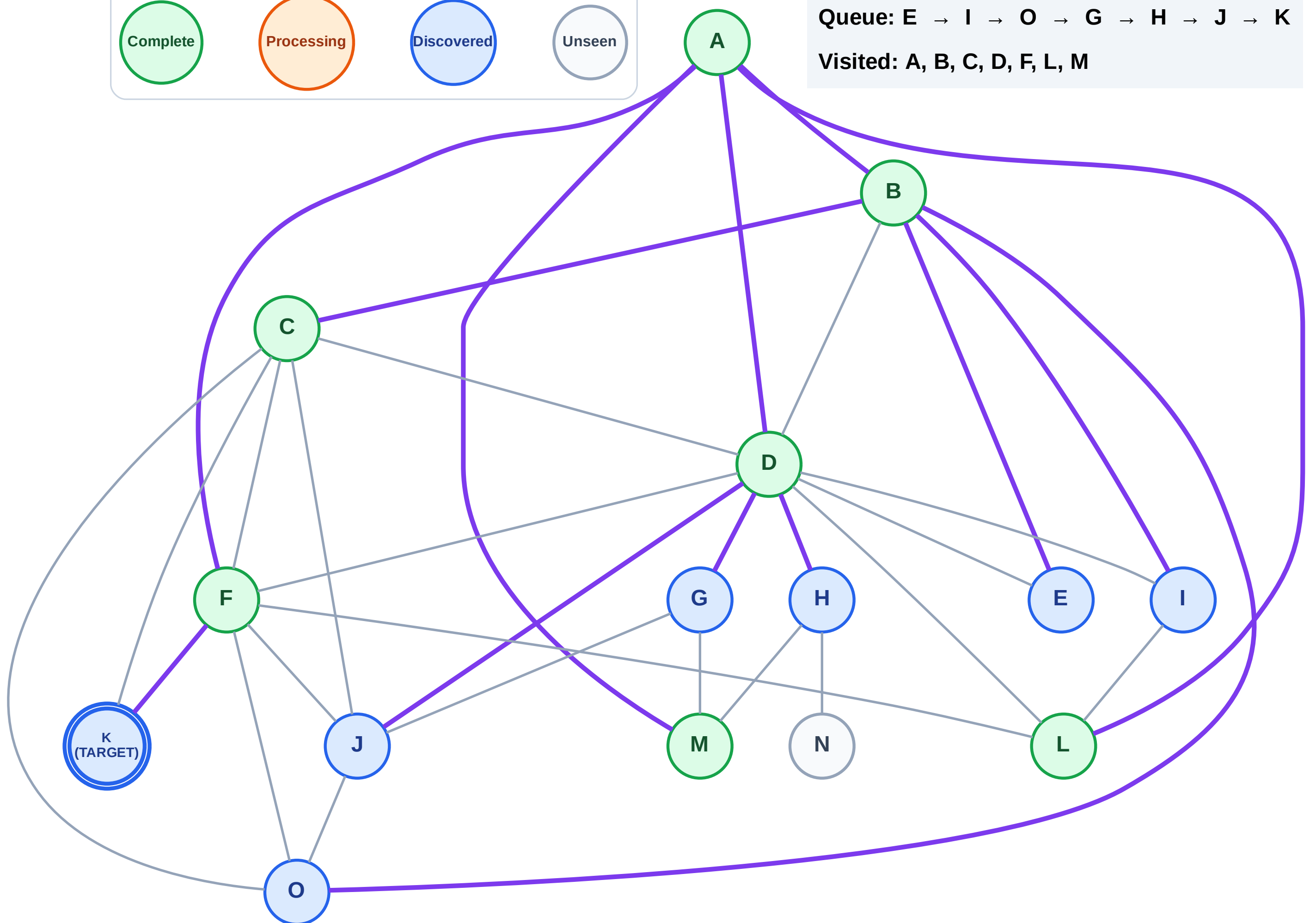
Step 15: No new neighbors from C

Legend



Queue: E → I → O → G → H → J → K

Visited: A, B, C, D, F, L, M



BFS Traversal

Step 16: Process node E

Legend



Complete



Processing



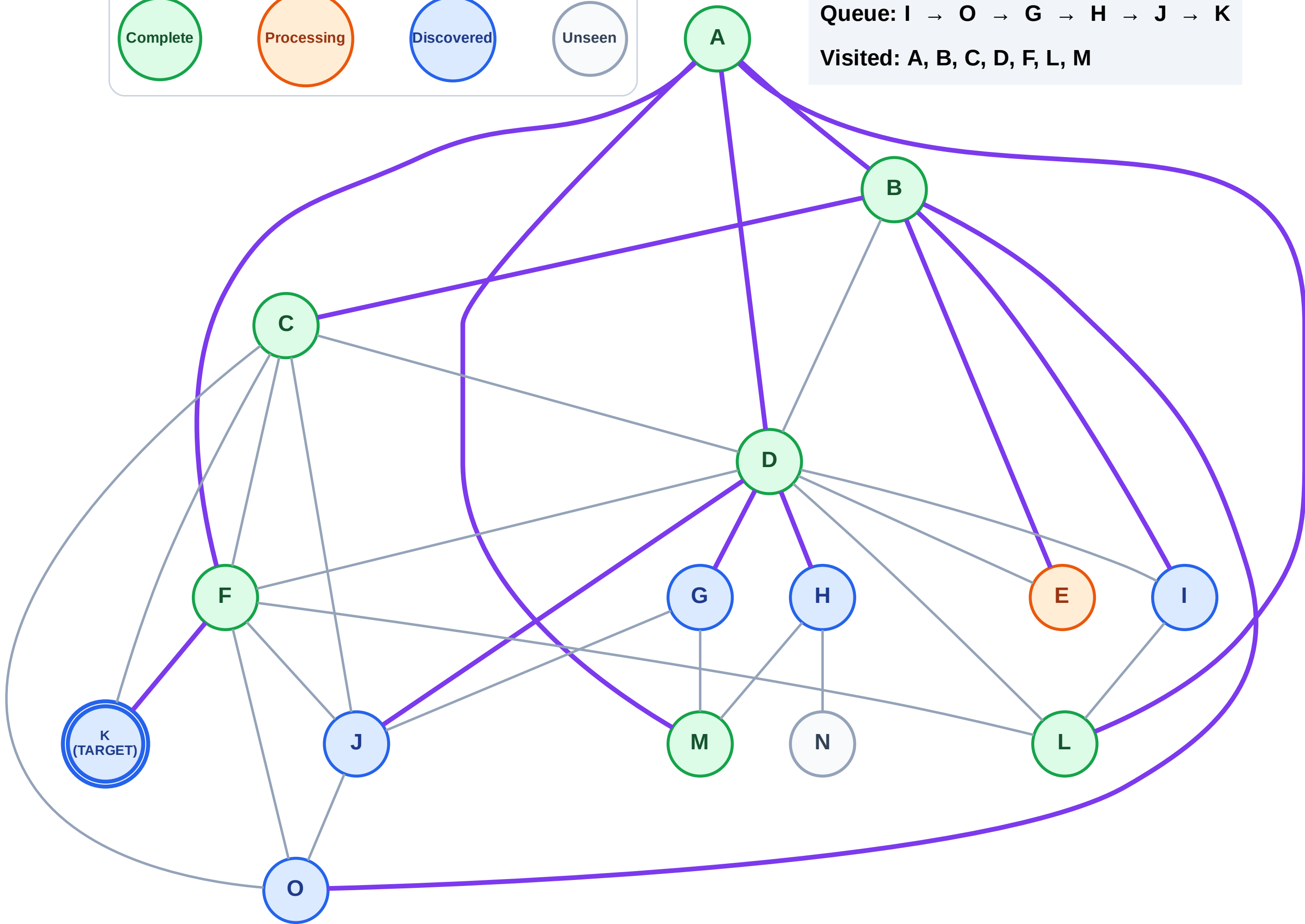
Discovered



Unseen

Queue: I → O → G → H → J → K

Visited: A, B, C, D, F, L, M

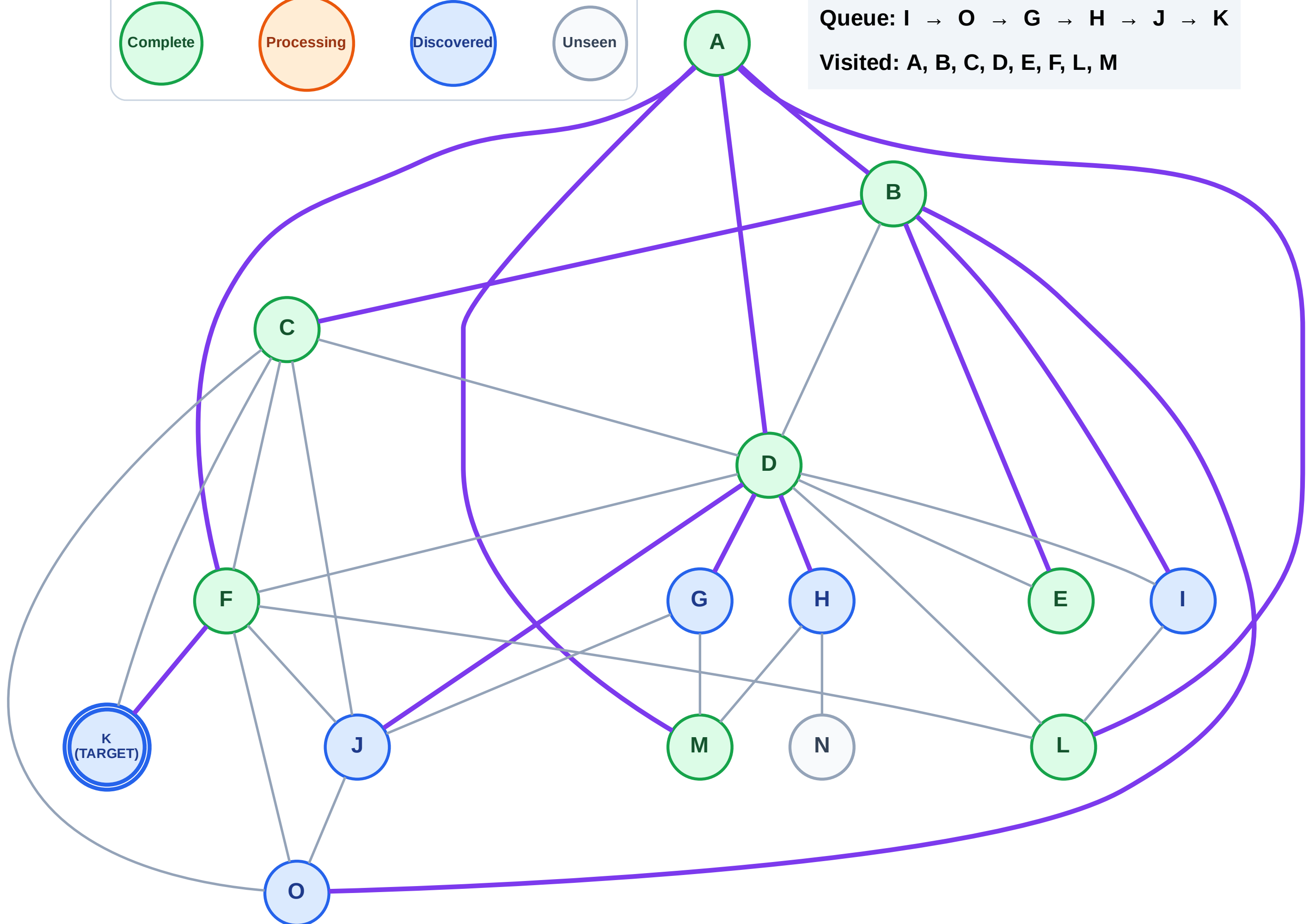


Step 17: No new neighbors from E

Legend

Unseen

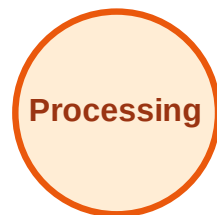
Visited: A, B, C, D, E, F, L, M



BFS Traversal

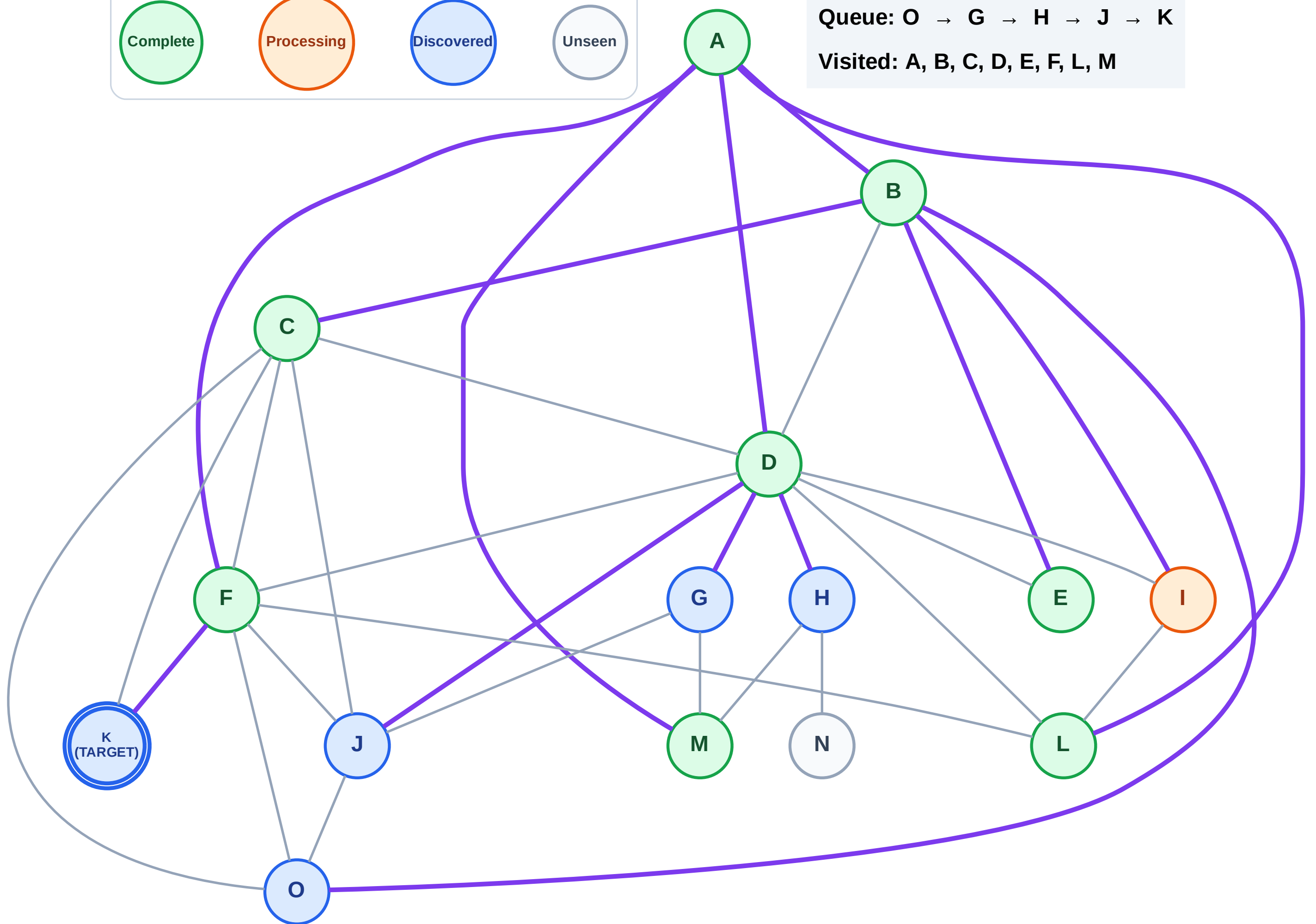
Step 18: Process node I

Legend



Queue: O → G → H → J → K

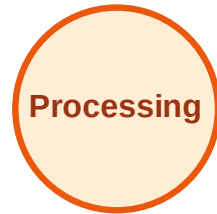
Visited: A, B, C, D, E, F, L, M



BFS Traversal

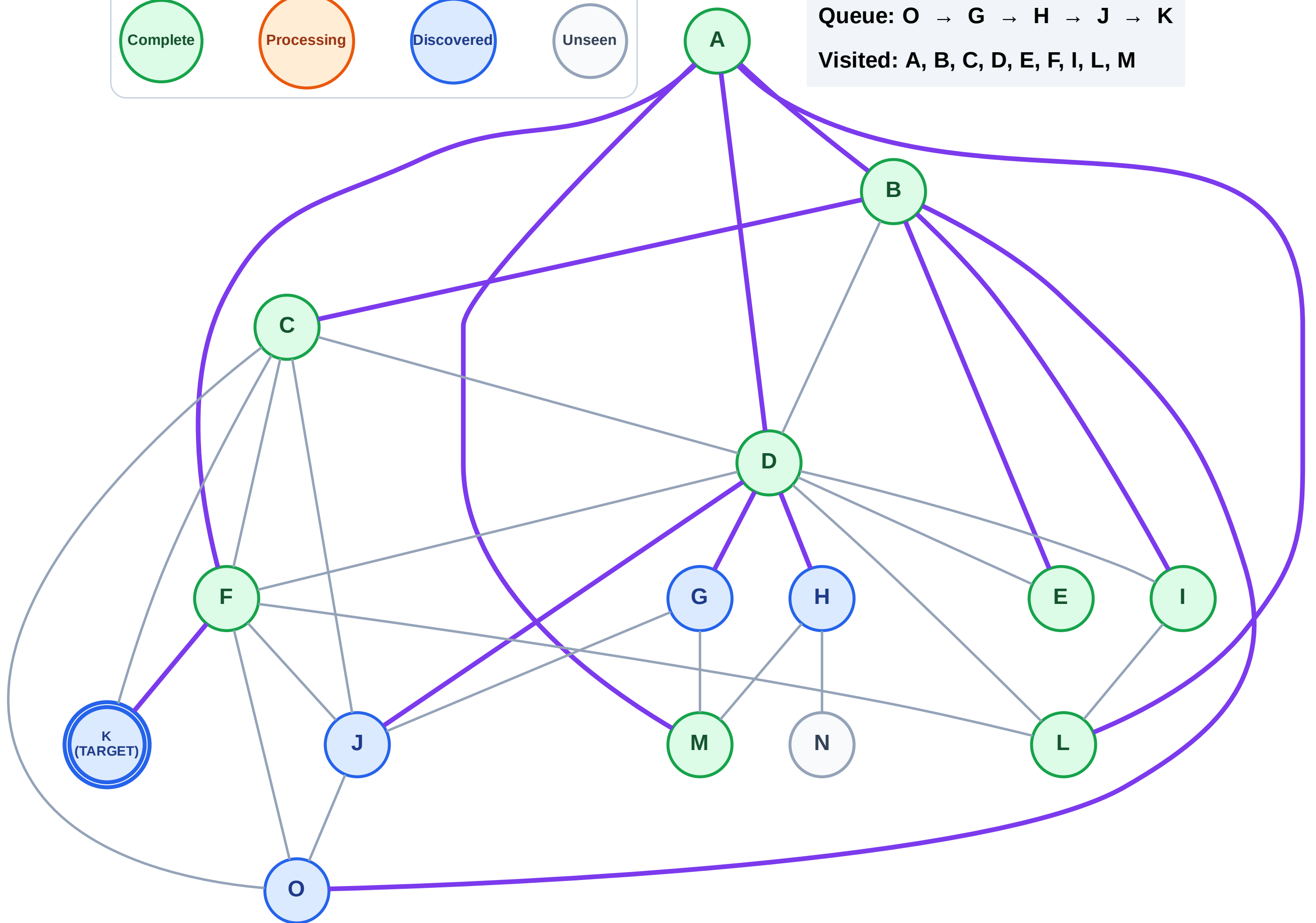
Step 19: No new neighbors from I

Legend



Queue: O → G → H → J → K

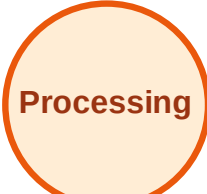
Visited: A, B, C, D, E, F, I, L, M



BFS Traversal

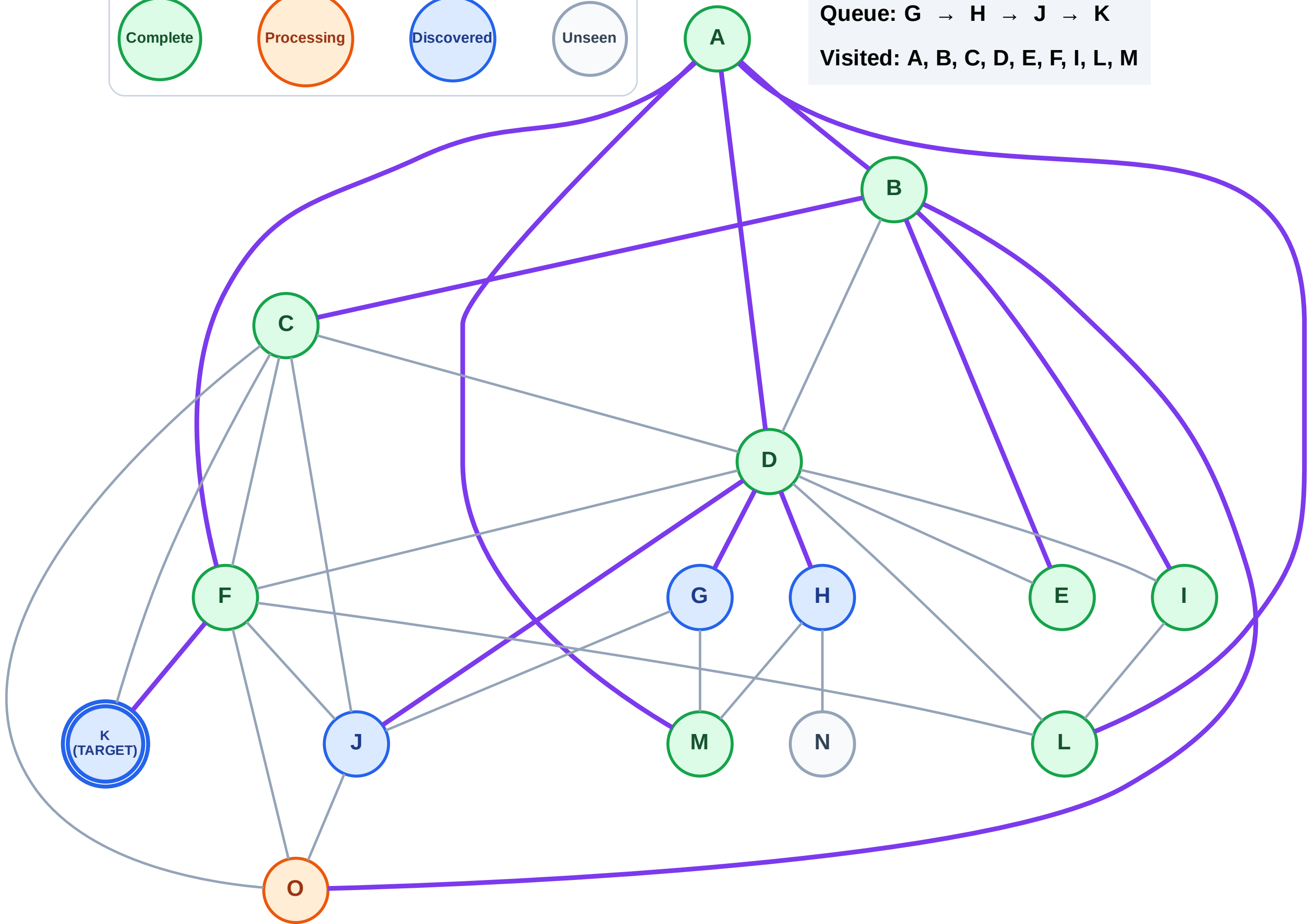
Step 20: Process node O

Legend



Queue: G → H → J → K

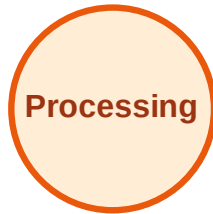
Visited: A, B, C, D, E, F, I, L, M



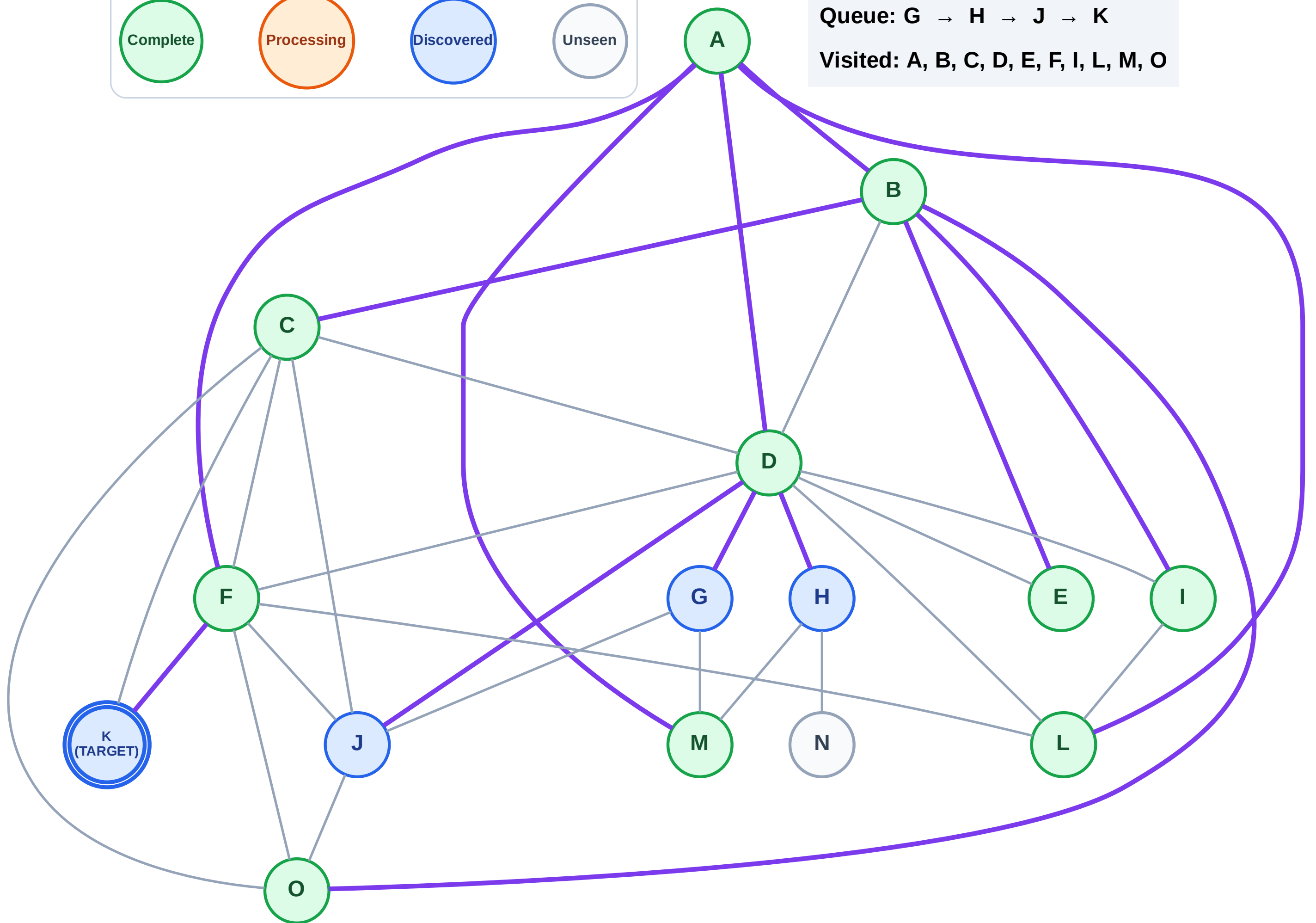
BFS Traversal

Step 21: No new neighbors from O

Legend



Queue: G → H → J → K
Visited: A, B, C, D, E, F, I, L, M, O



BFS Traversal

Step 22: Process node G

Legend



Complete



Processing



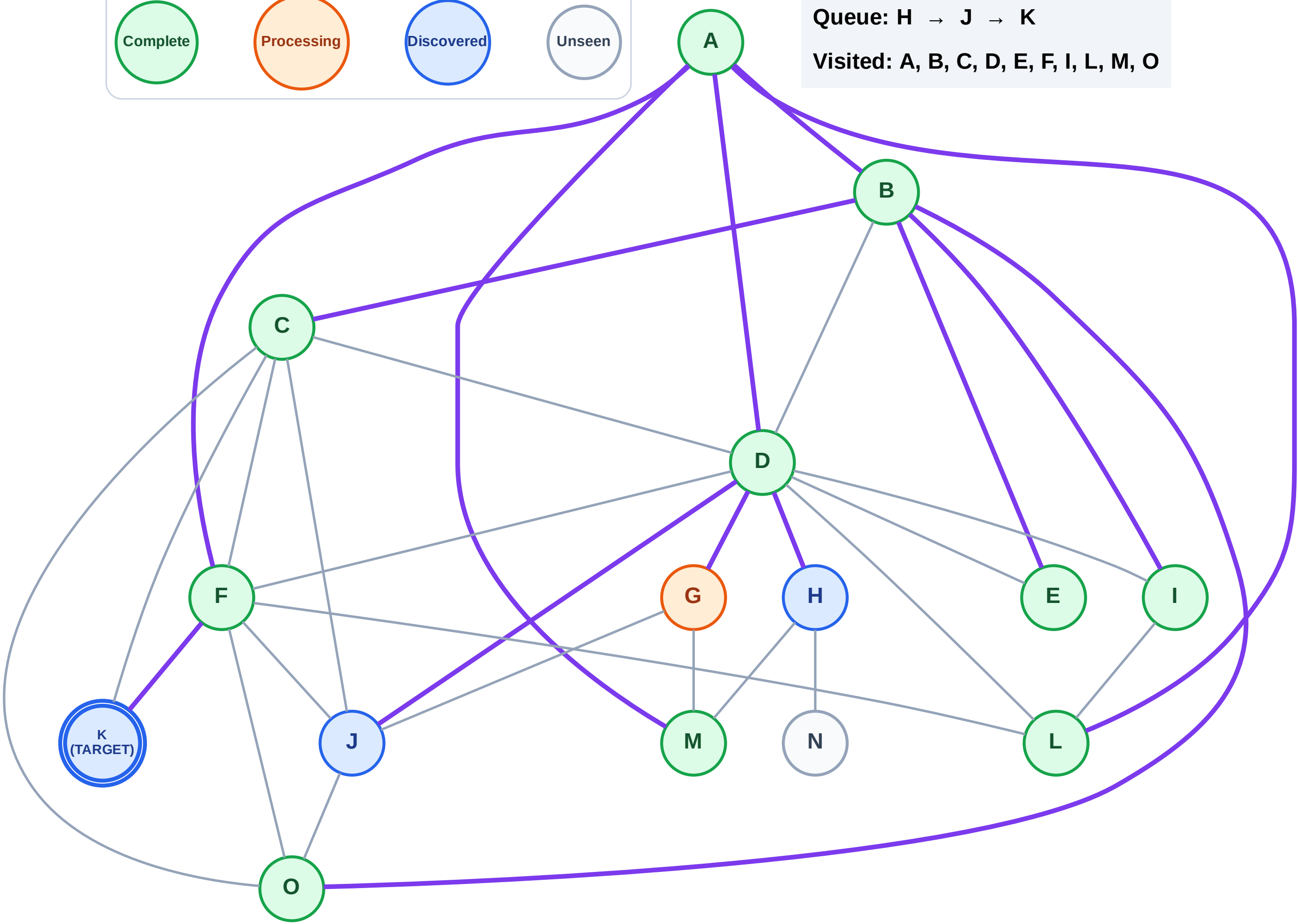
Discovered



Unseen

Queue: H → J → K

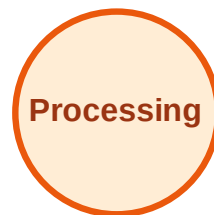
Visited: A, B, C, D, E, F, I, L, M, O



BFS Traversal

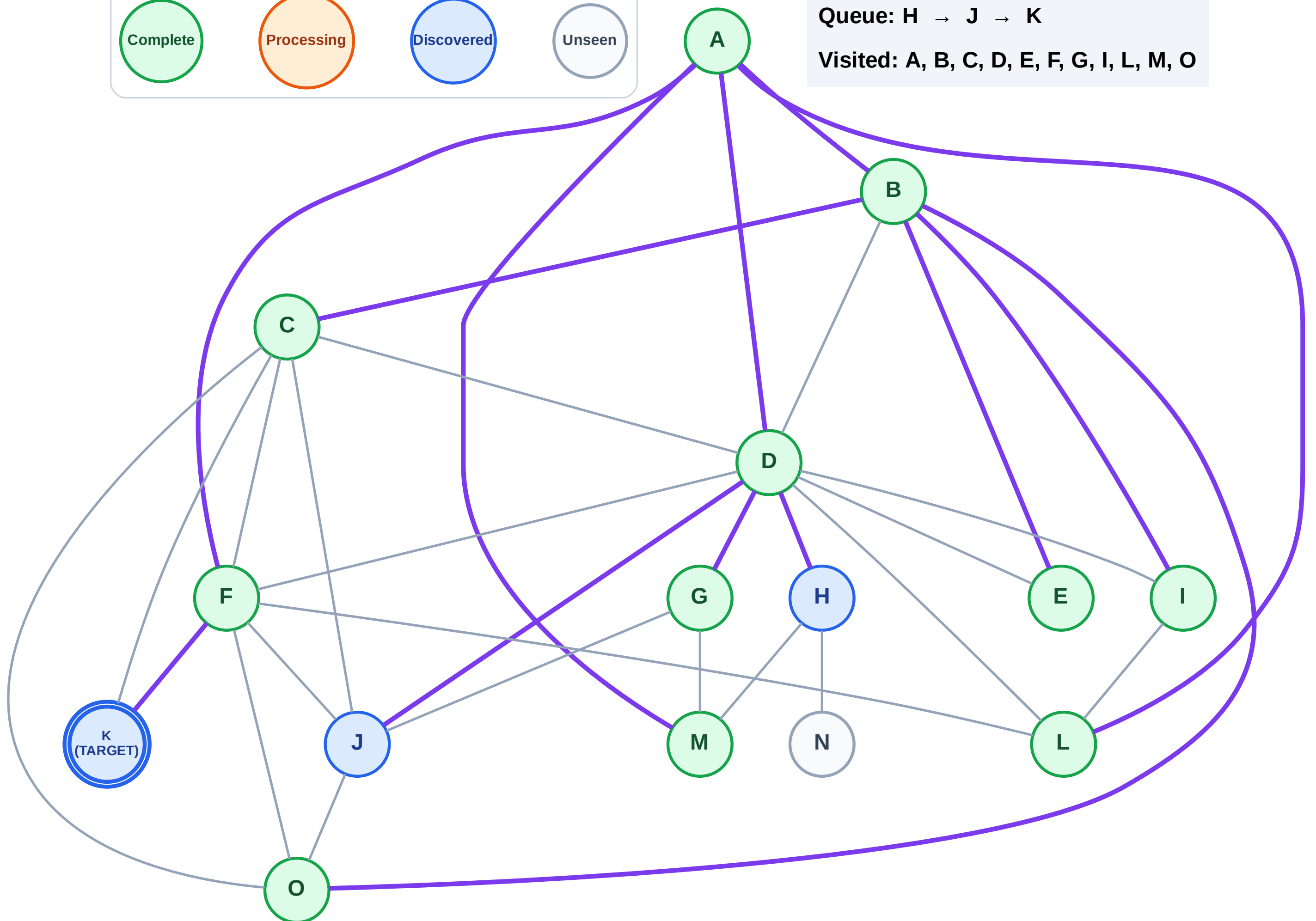
Step 23: No new neighbors from G

Legend



Queue: H → J → K

Visited: A, B, C, D, E, F, G, I, L, M, O



BFS Traversal

Step 24: Process node H

Legend

Complete

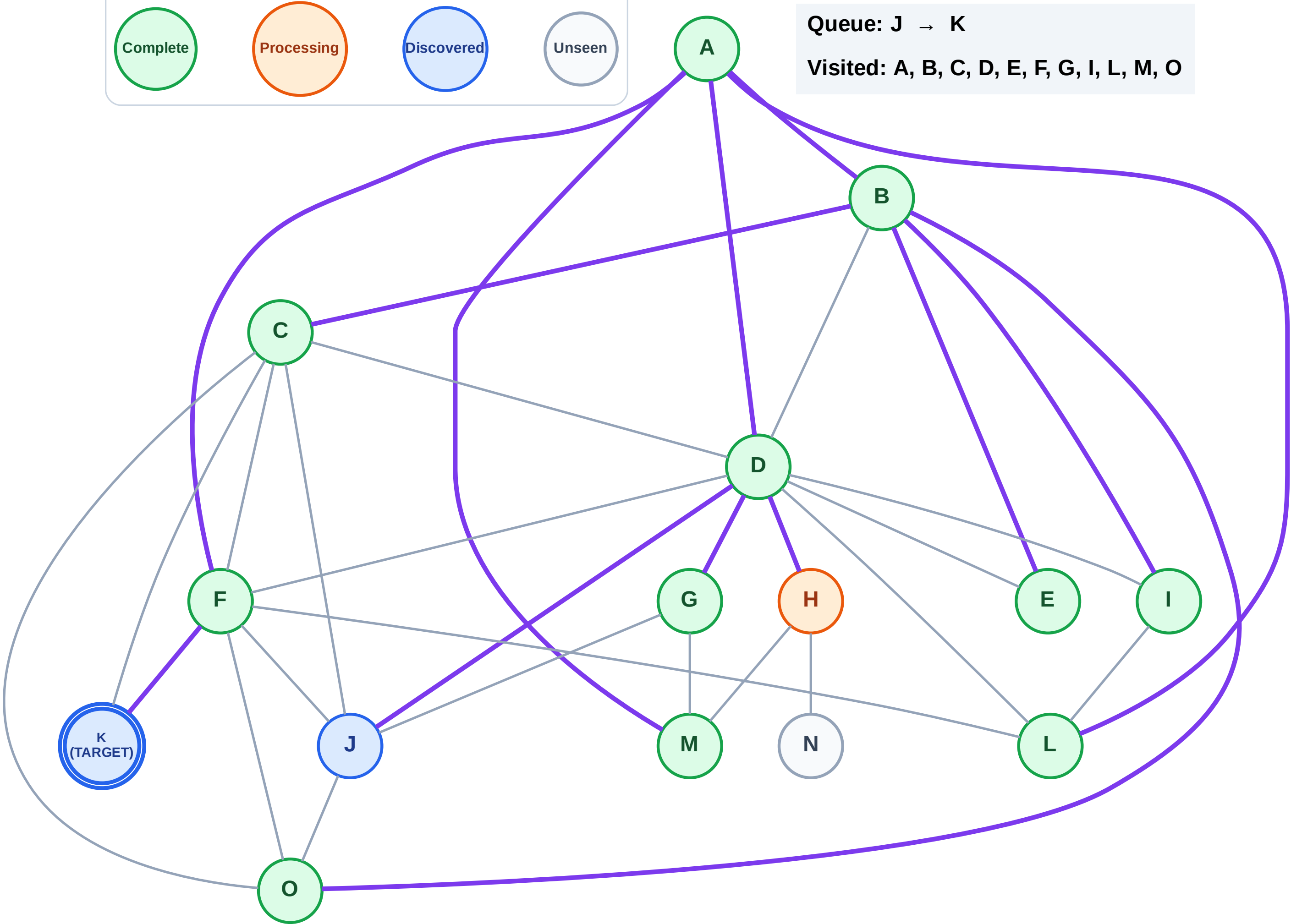
Processing

Discovered

Unseen

Queue: J → K

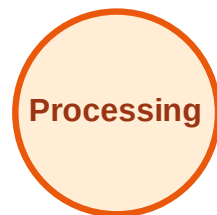
Visited: A, B, C, D, E, F, G, I, L, M, O



BFS Traversal

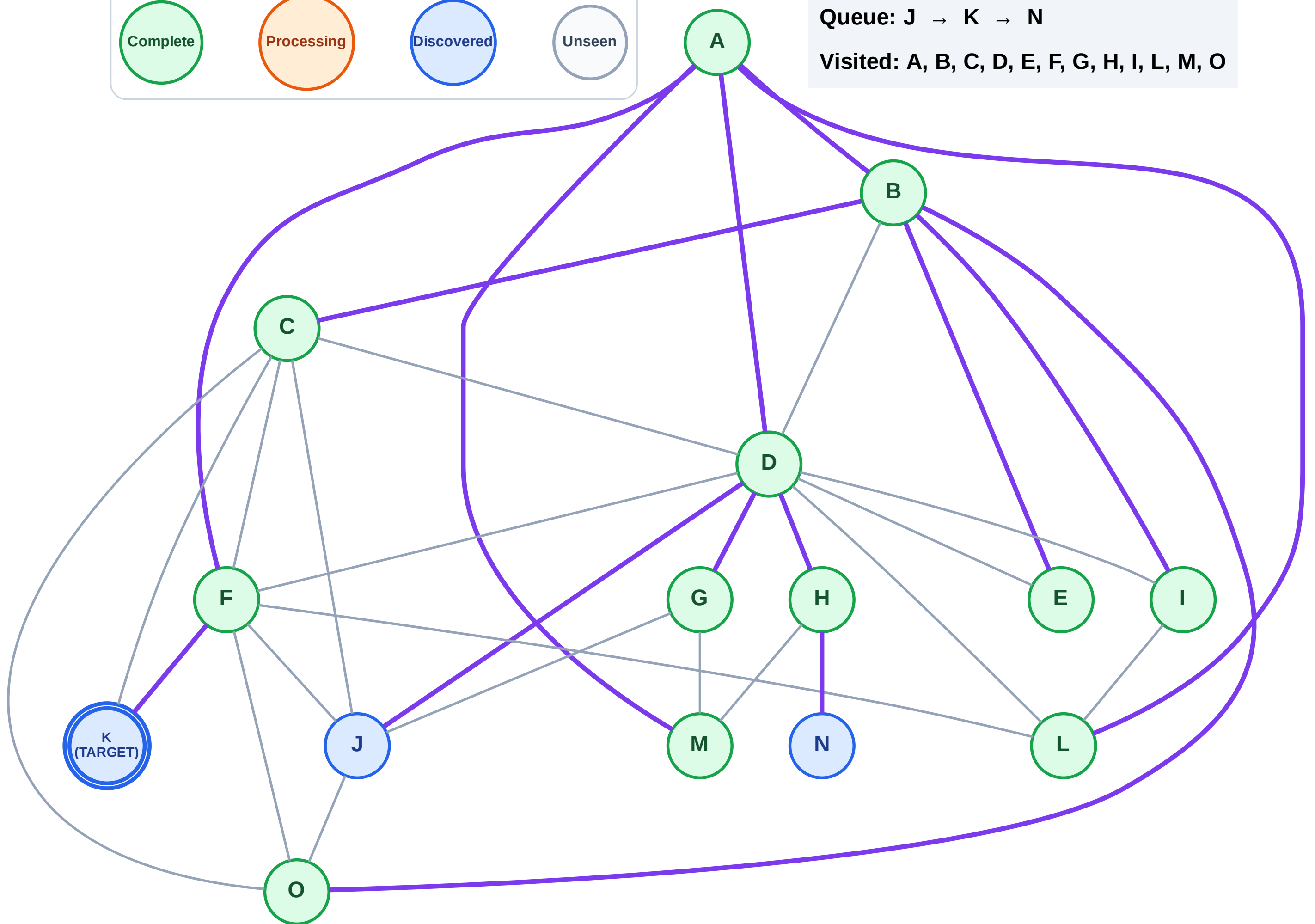
Step 25: Discover N from H

Legend



Queue: J → K → N

Visited: A, B, C, D, E, F, G, H, I, L, M, O



BFS Traversal

Step 26: Process node J

Legend

Complete

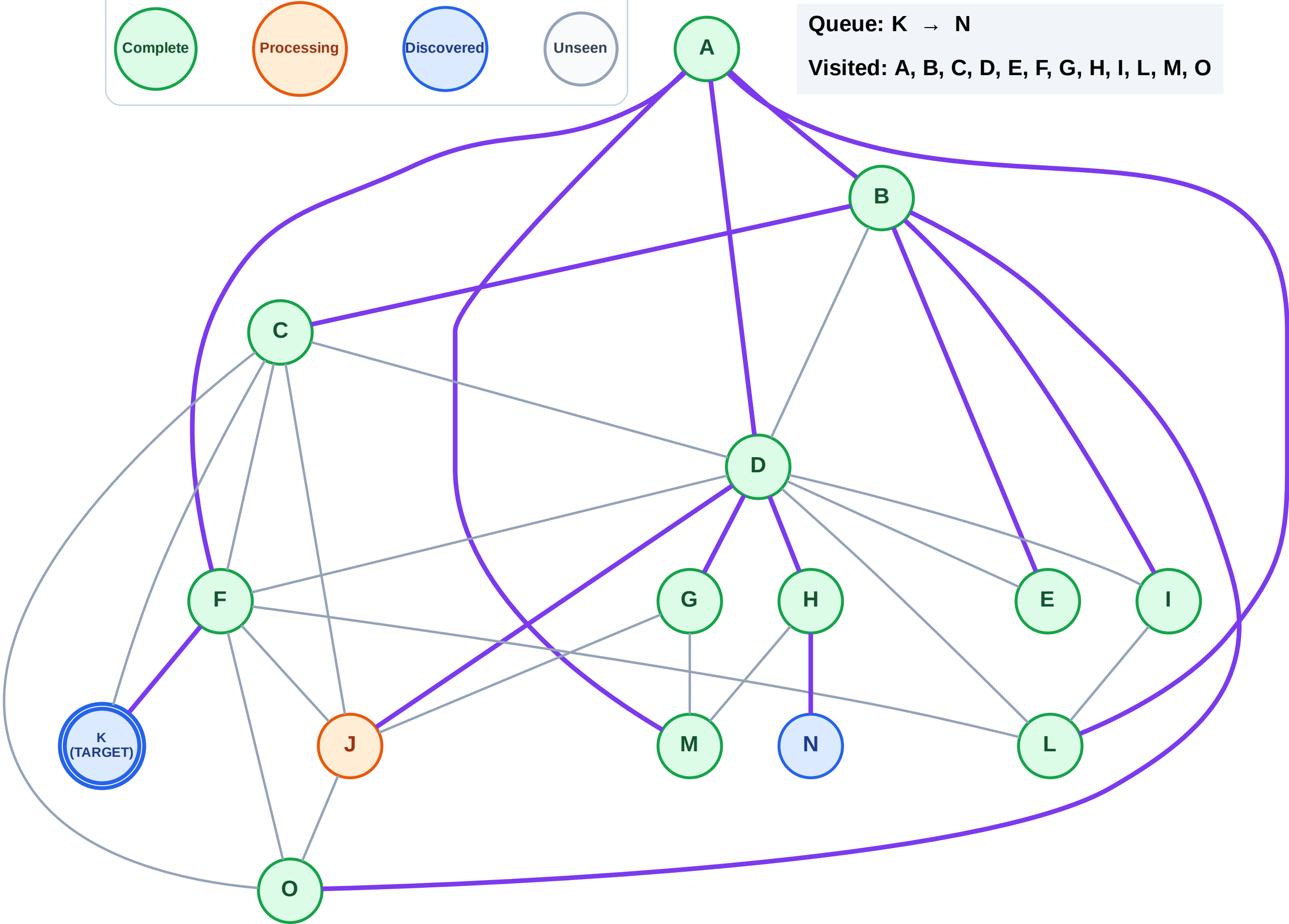
Processing

Discovered

Unseen

Queue: K → N

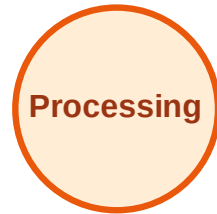
Visited: A, B, C, D, E, F, G, H, I, L, M, O



BFS Traversal

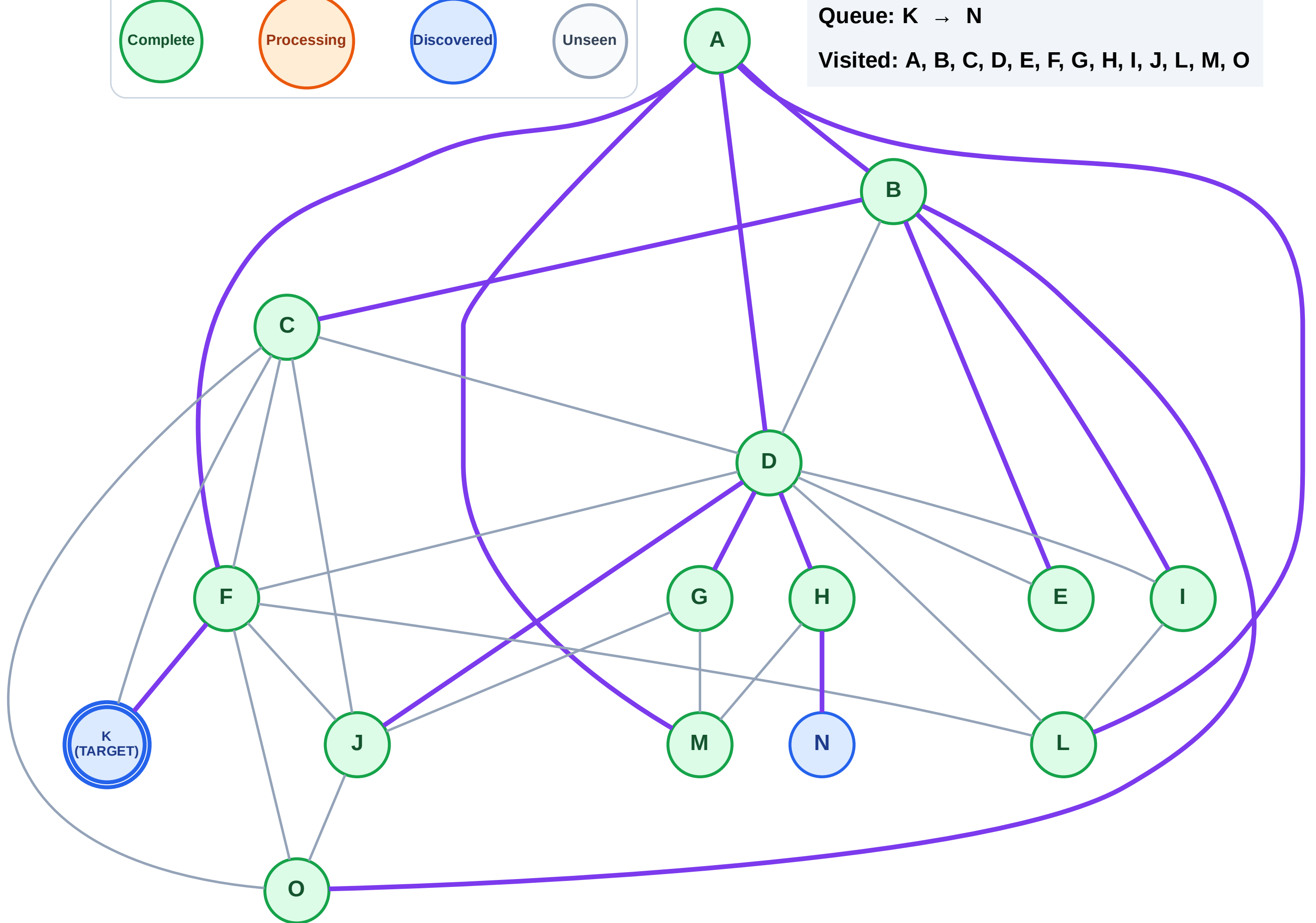
Step 27: No new neighbors from J

Legend



Queue: K → N

Visited: A, B, C, D, E, F, G, H, I, J, L, M, O



BFS Traversal

Step 28: Process node K

Legend

Complete

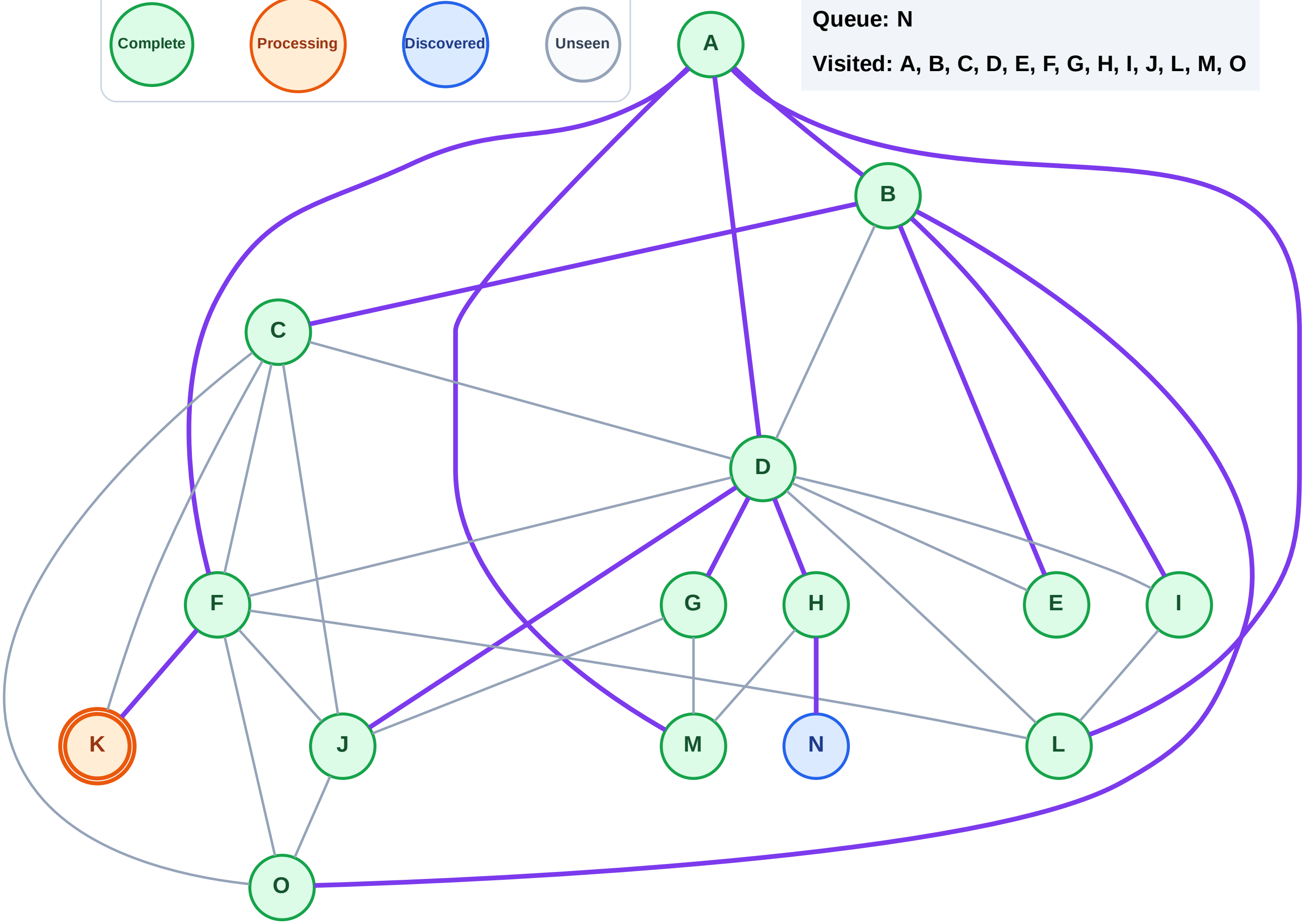
Processing

Discovered

Unseen

Queue: N

Visited: A, B, C, D, E, F, G, H, I, J, L, M, O



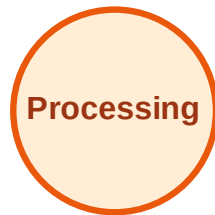
BFS Traversal

Step 29: TARGET NODE K REACHED! Path (2 hops): A → F → K

Legend



Complete



Processing



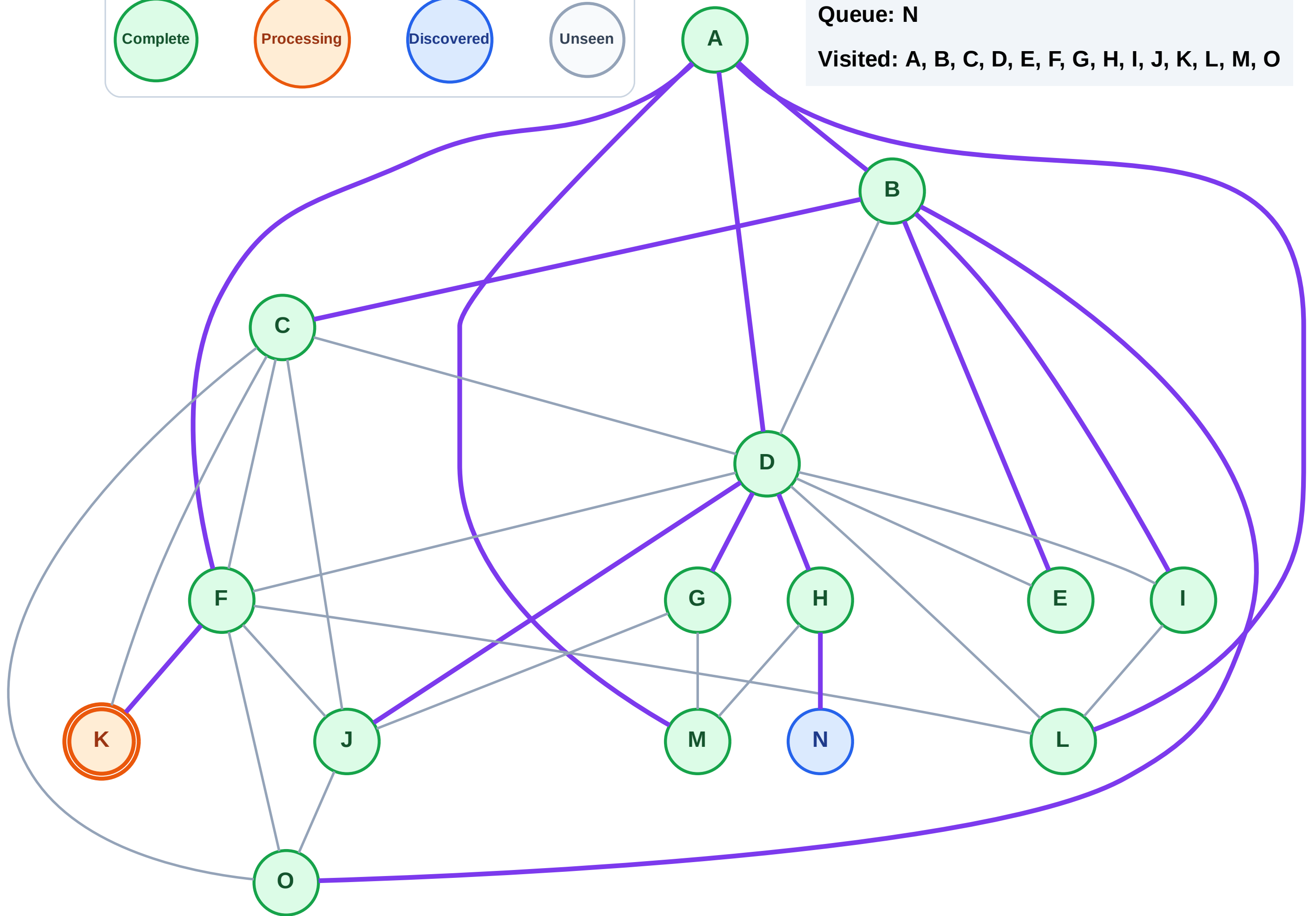
Discovered



Unseen

Queue: N

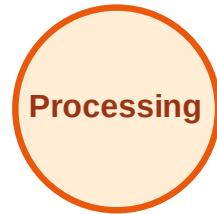
Visited: A, B, C, D, E, F, G, H, I, J, K, L, M, O



DFS Traversal

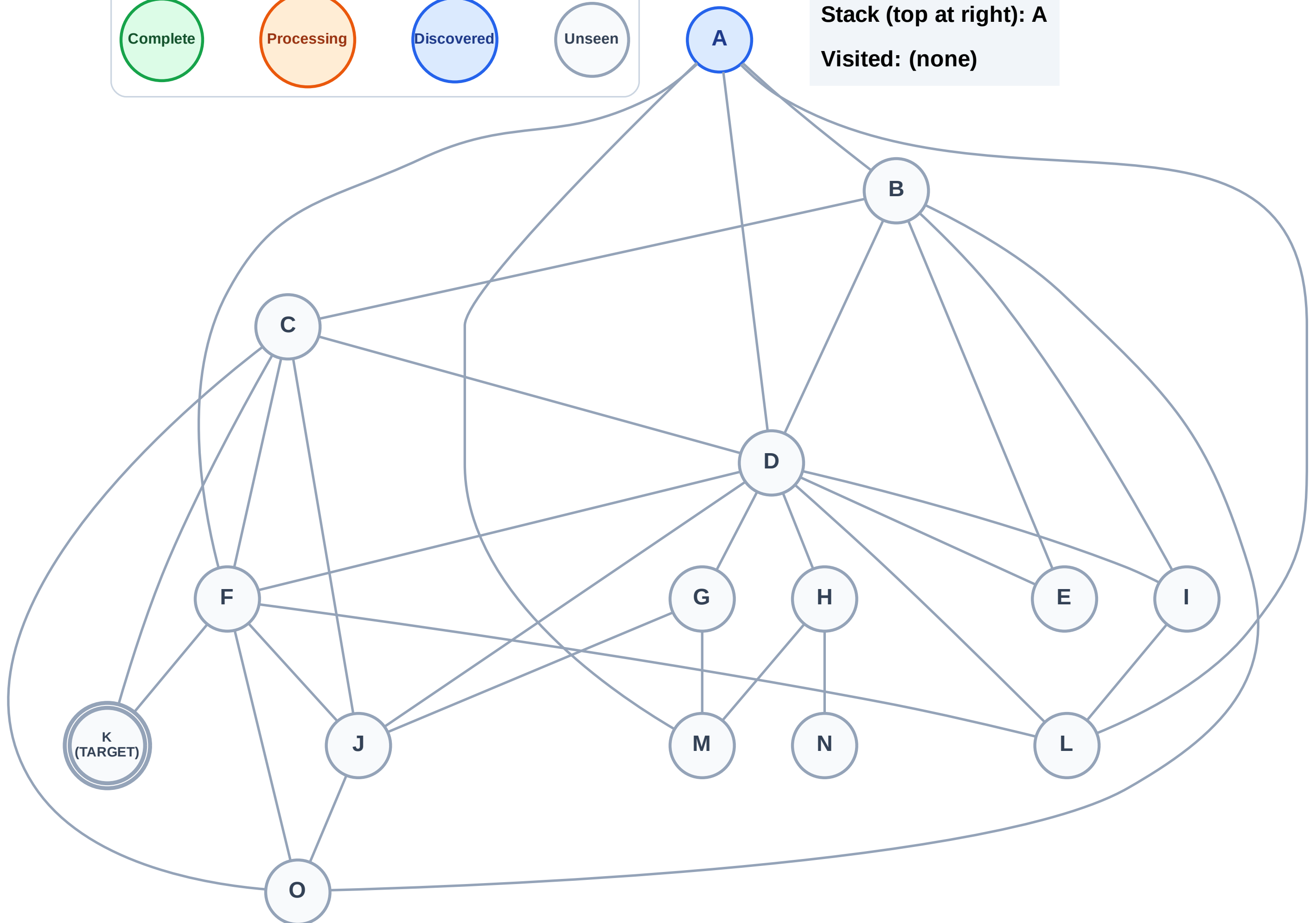
Step 1: Start at A (Seeking target node K)

Legend



Stack (top at right): A

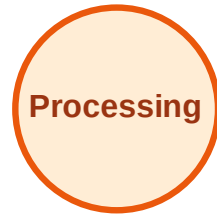
Visited: (none)



DFS Traversal

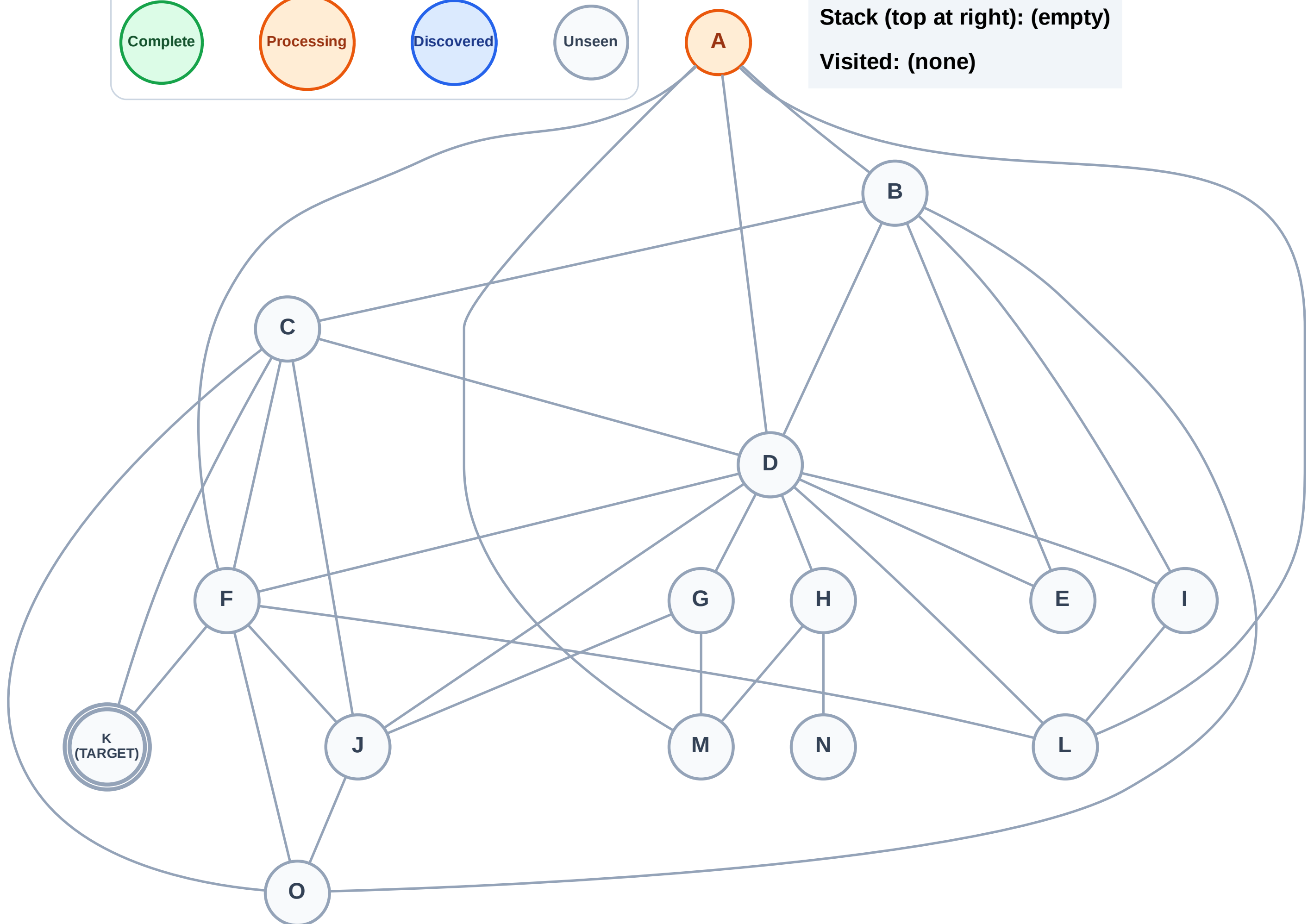
Step 2: Process node A

Legend



Stack (top at right): (empty)

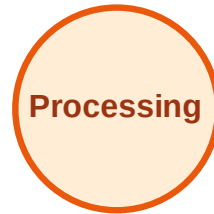
Visited: (none)



DFS Traversal

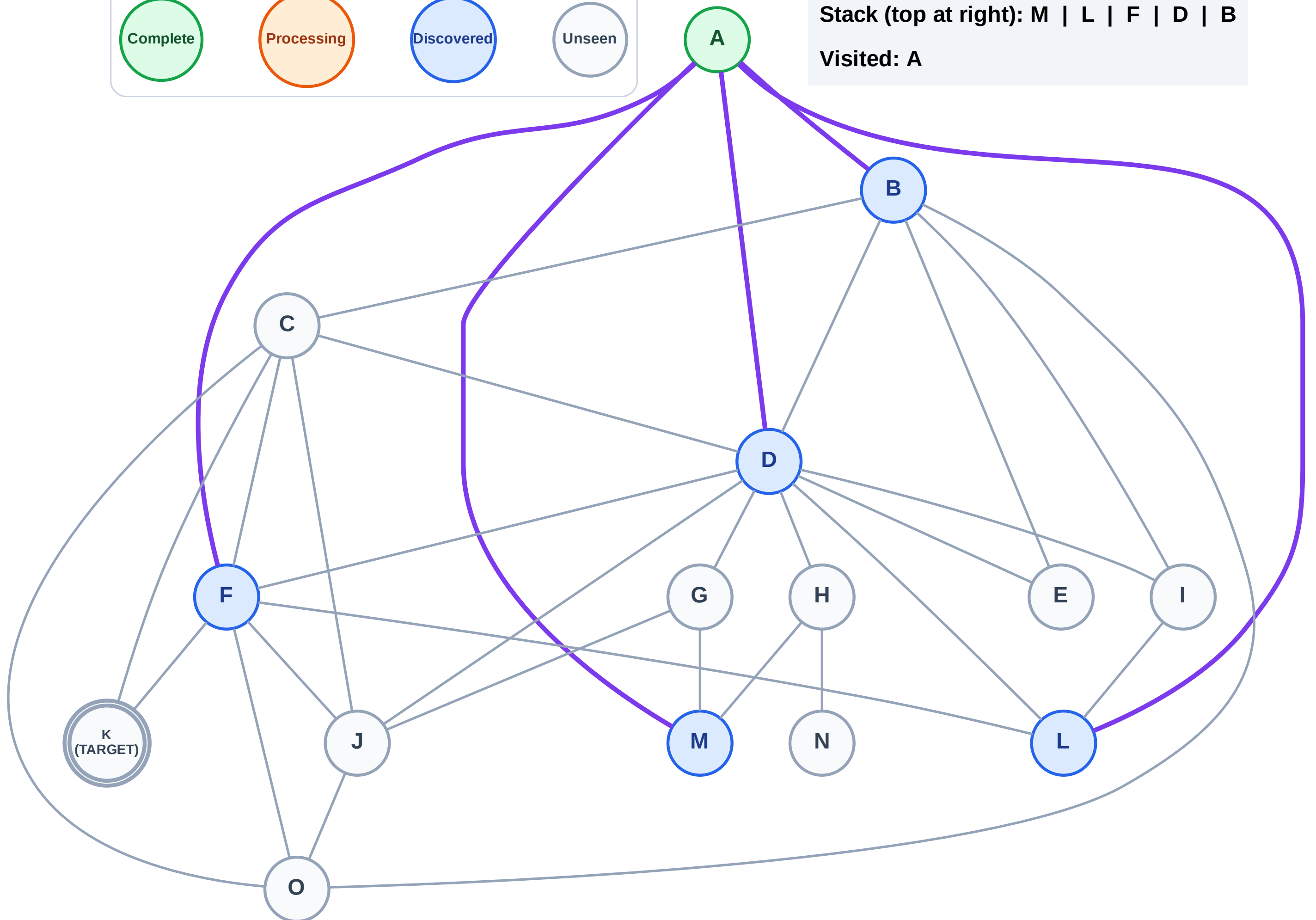
Step 3: Discover M, L, F, D, B from A

Legend



Stack (top at right): M | L | F | D | B

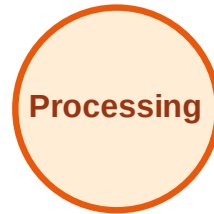
Visited: A



DFS Traversal

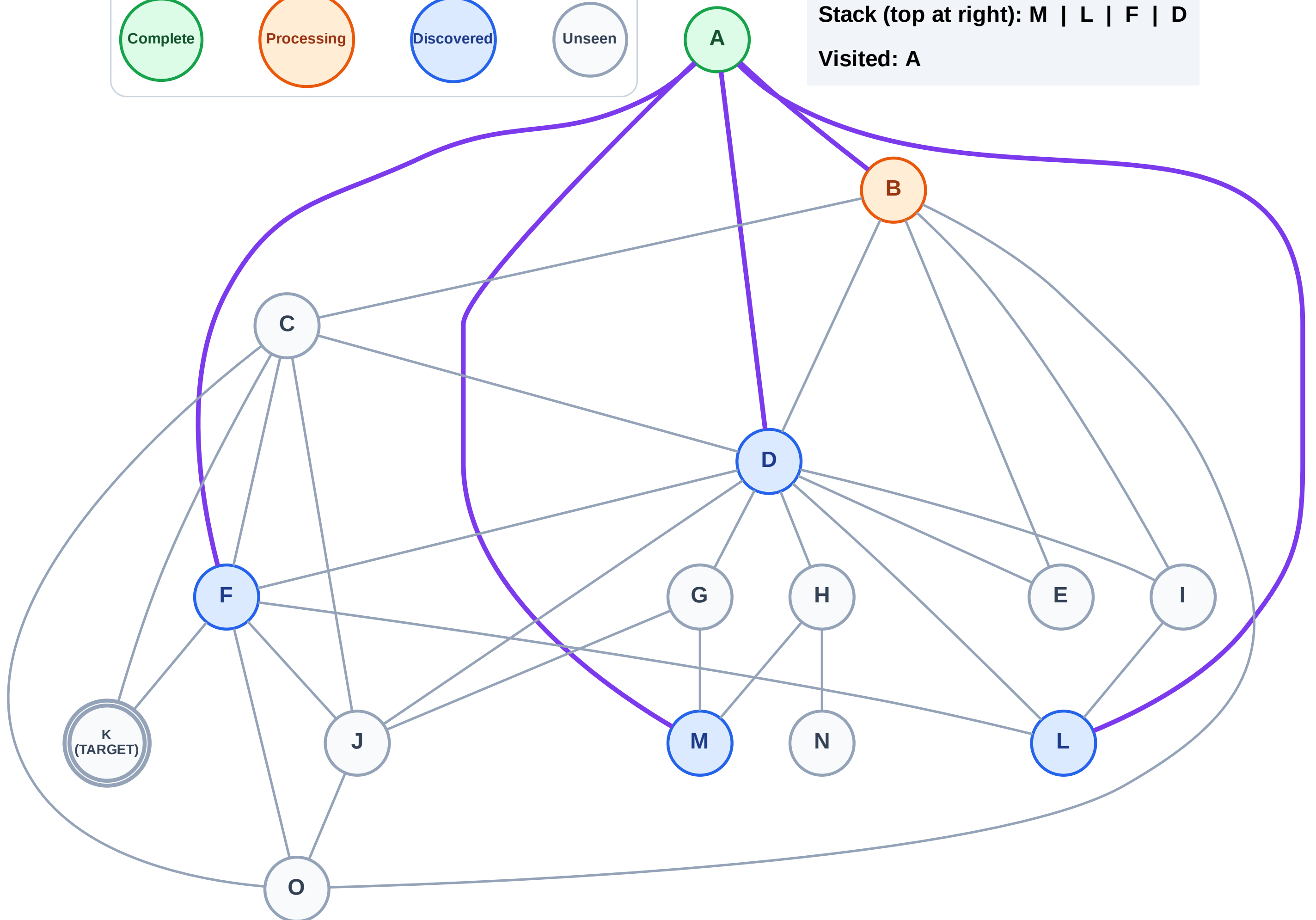
Step 4: Process node B

Legend



Stack (top at right): M | L | F | D

Visited: A



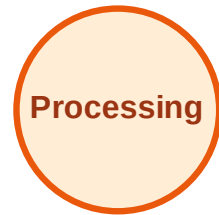
DFS Traversal

Step 5: Discover O, I, E, C from B

Legend



Complete



Processing



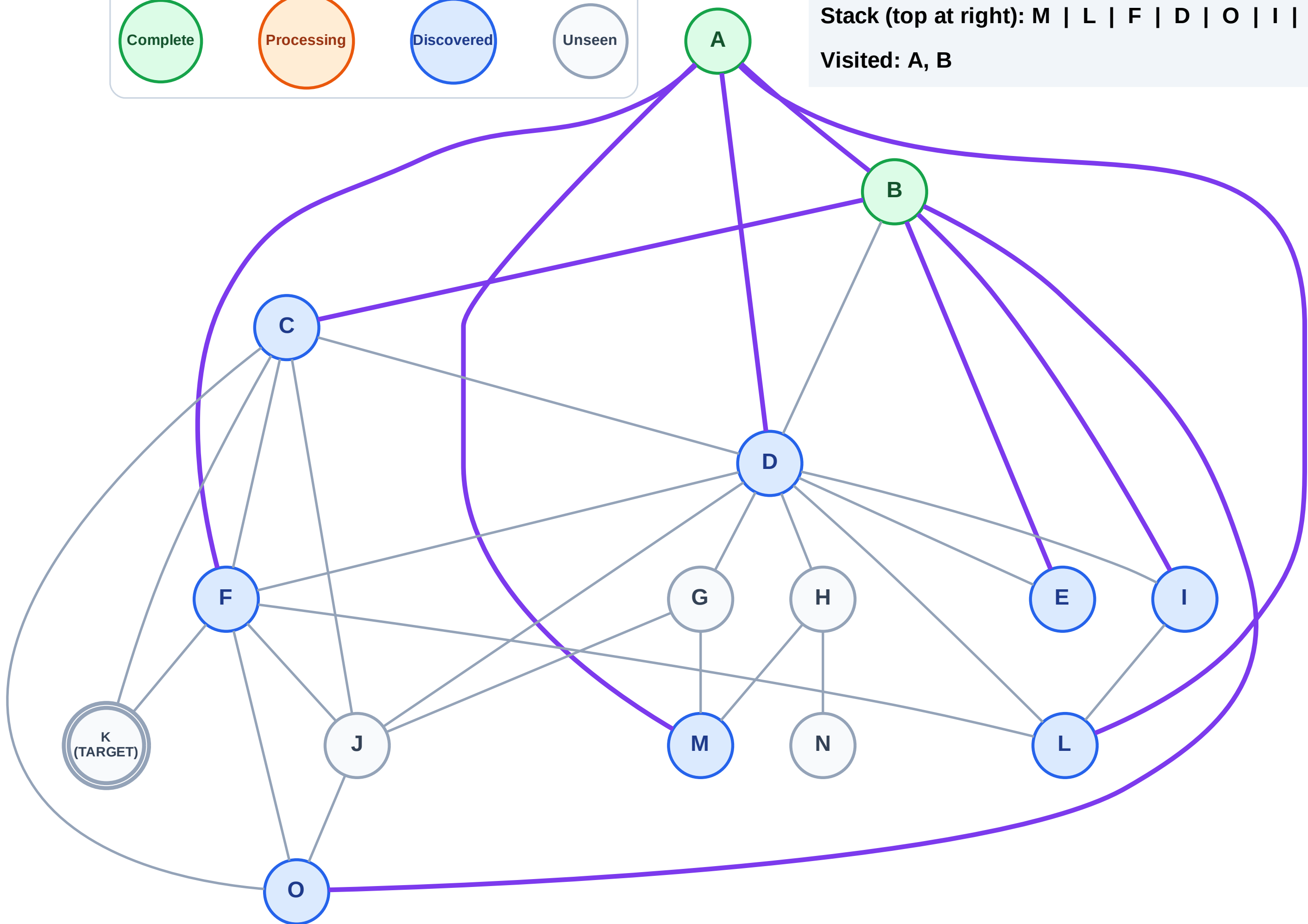
Discovered



Unseen

Stack (top at right): M | L | F | D | O | I | E | C

Visited: A, B



DFS Traversal

Step 6: Process node C

Legend

Complete

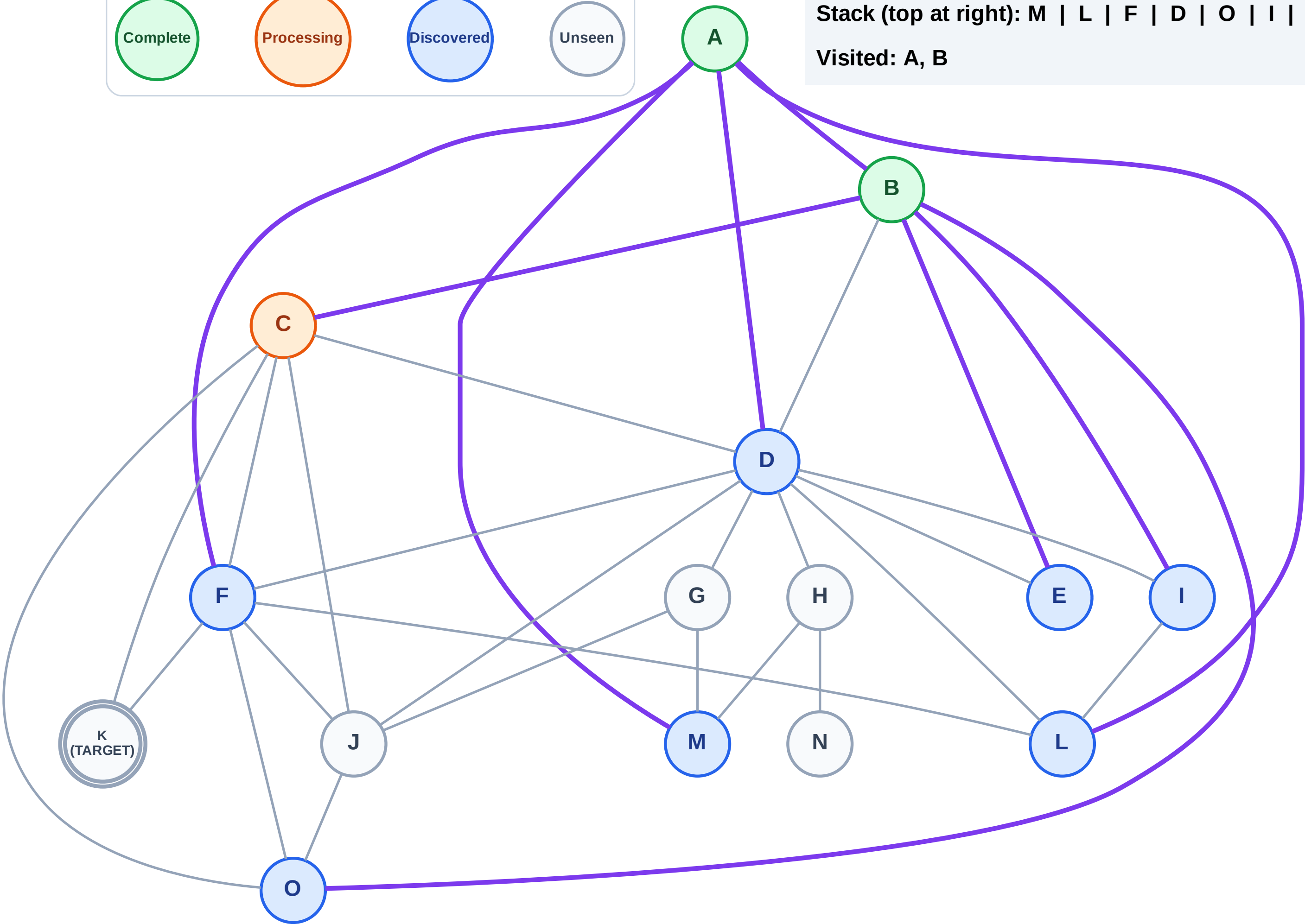
Processing

Discovered

Unseen

Stack (top at right): M | L | F | D | O | I | E

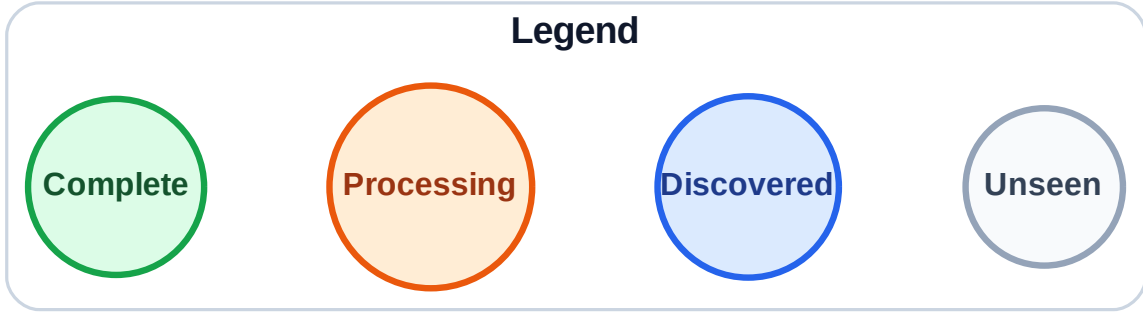
Visited: A, B



Step 7: Discover K, J from C

Stack (top at right): M | L | F | D | O | I | E | K | J

Visited: A, B, C



DFS Traversal

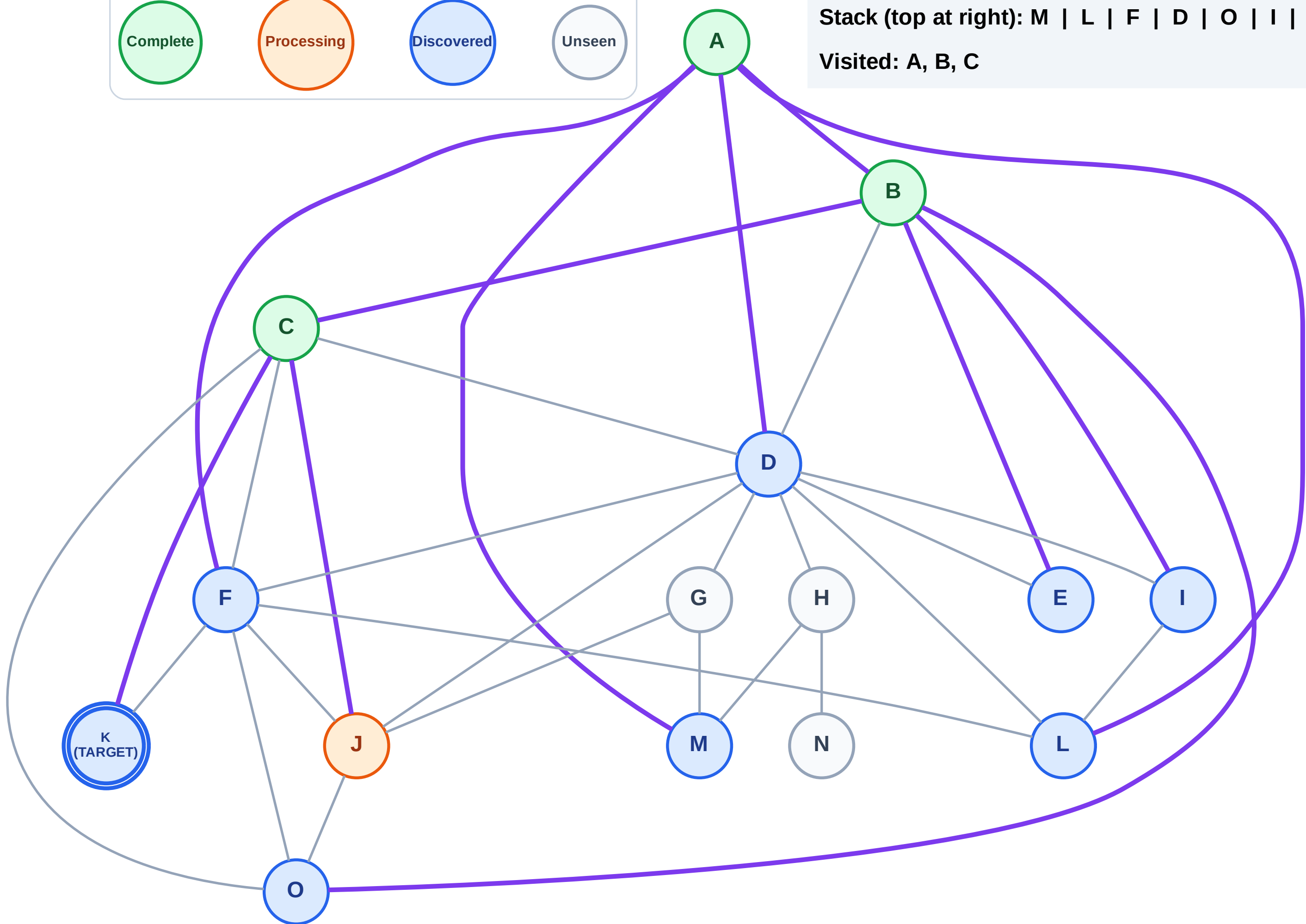
Step 8: Process node J

Legend



Stack (top at right): M | L | F | D | O | I | E | K

Visited: A, B, C



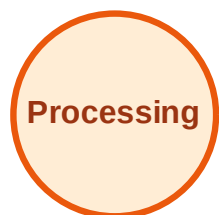
DFS Traversal

Step 9: Discover G from J

Legend



Complete



Processing



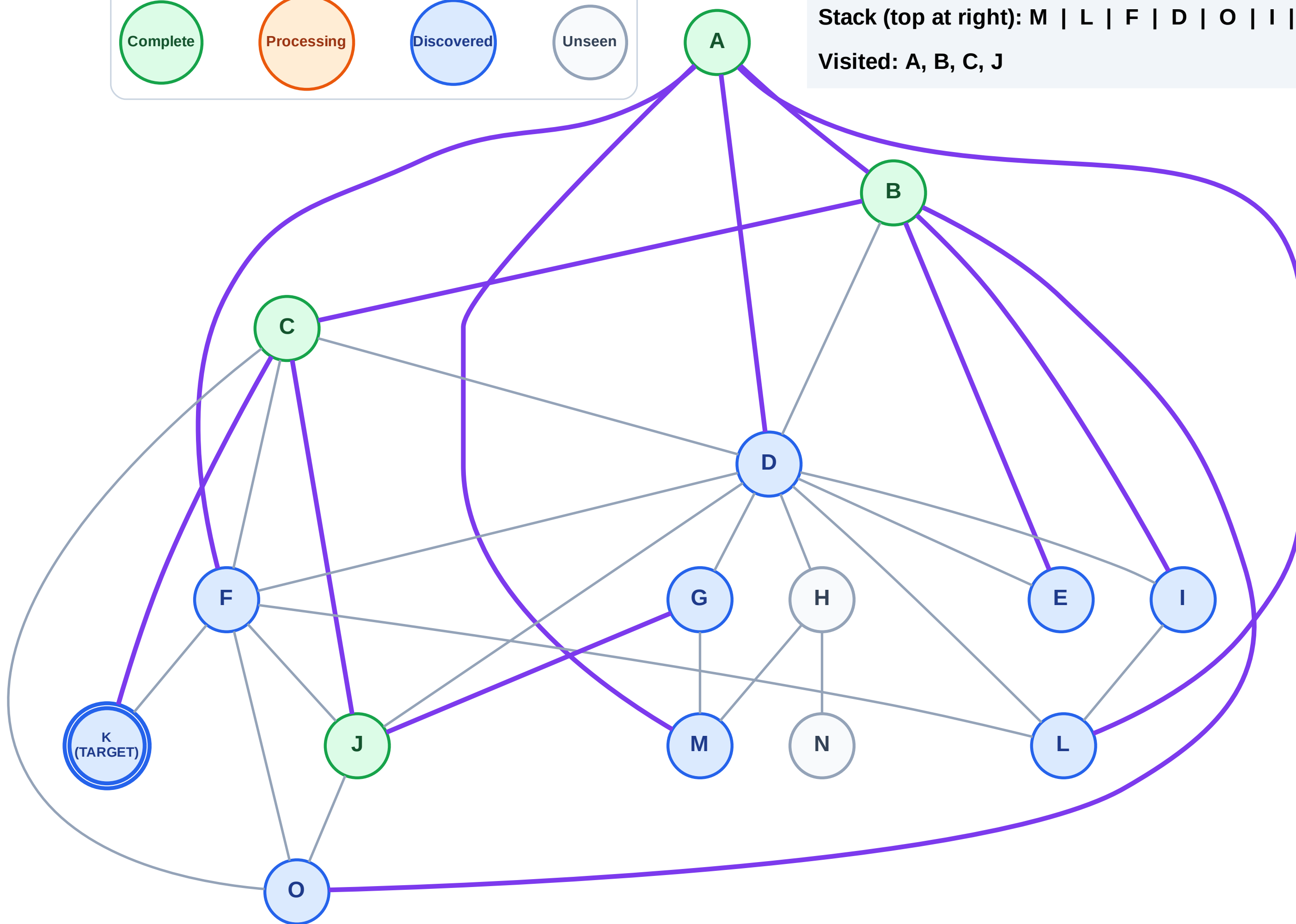
Discovered



Unseen

Stack (top at right): M | L | F | D | O | I | E | K | G

Visited: A, B, C, J



DFS Traversal

Step 10: Process node G

Legend

Complete

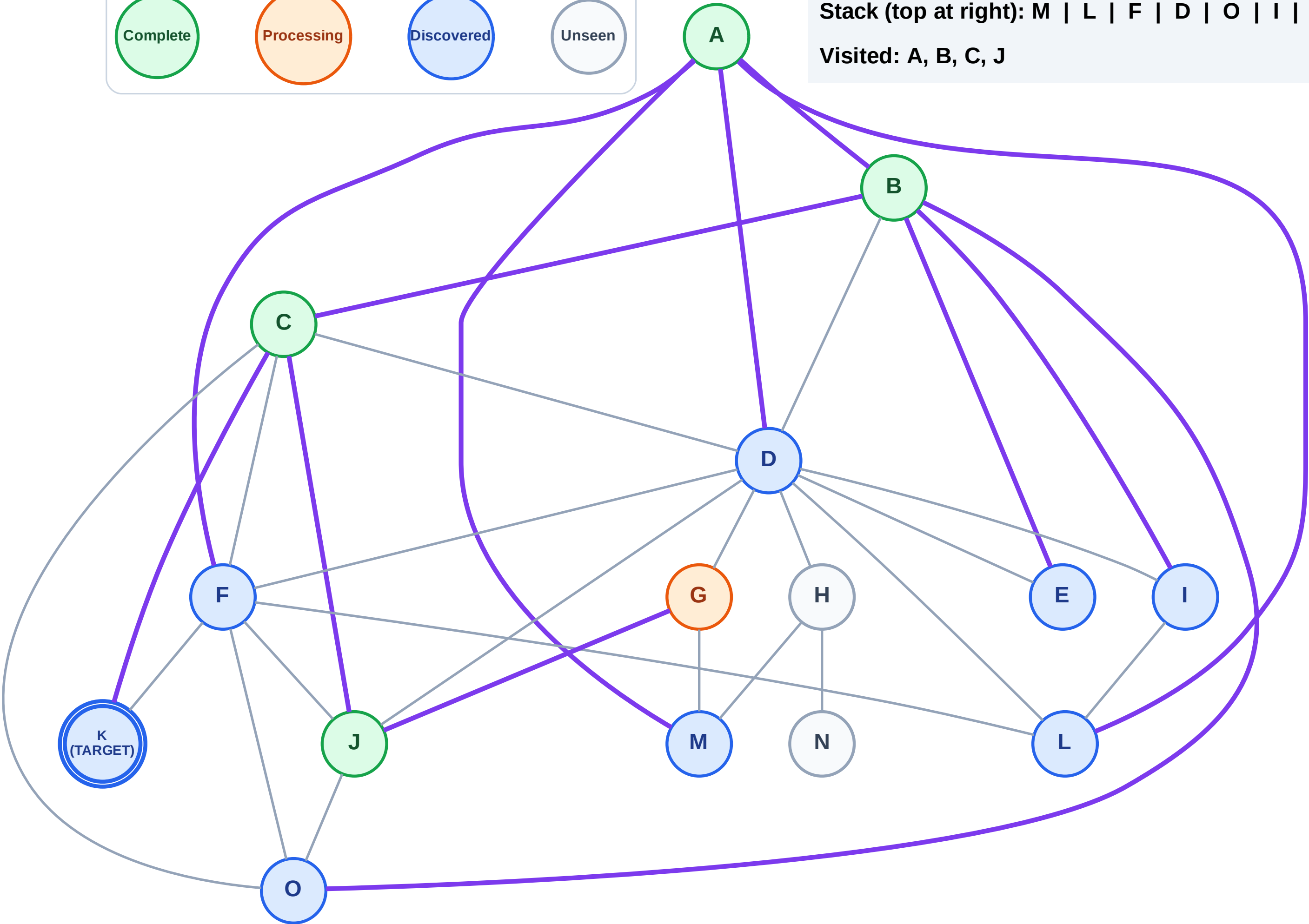
Processing

Discovered

Unseen

Stack (top at right): M | L | F | D | O | I | E | K

Visited: A, B, C, J



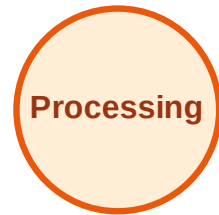
DFS Traversal

Step 11: No new neighbors from G

Legend



Complete



Processing



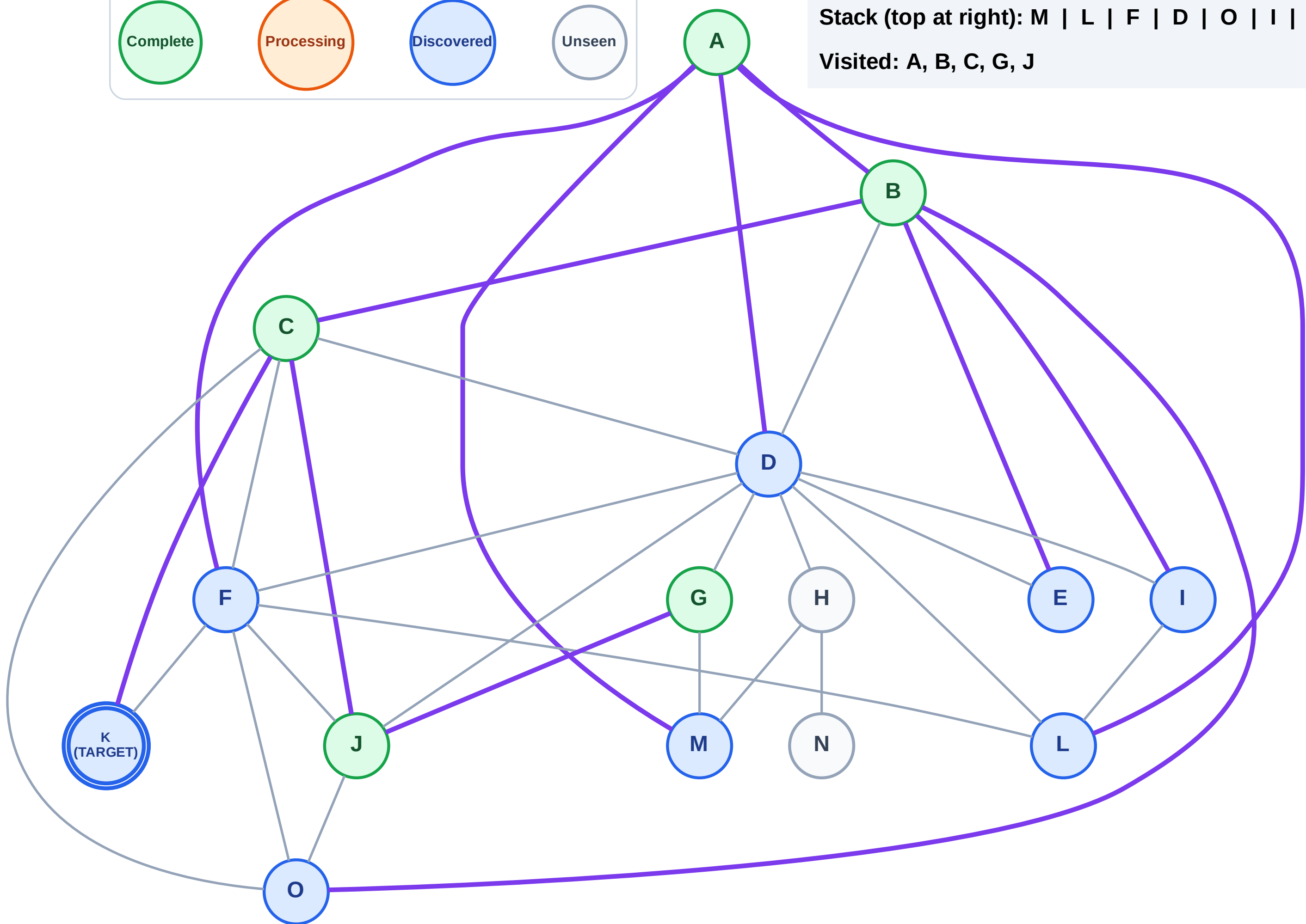
Discovered



Unseen

Stack (top at right): M | L | F | D | O | I | E | K

Visited: A, B, C, G, J



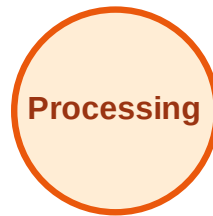
DFS Traversal

Step 12: Process node K

Legend



Complete



Processing



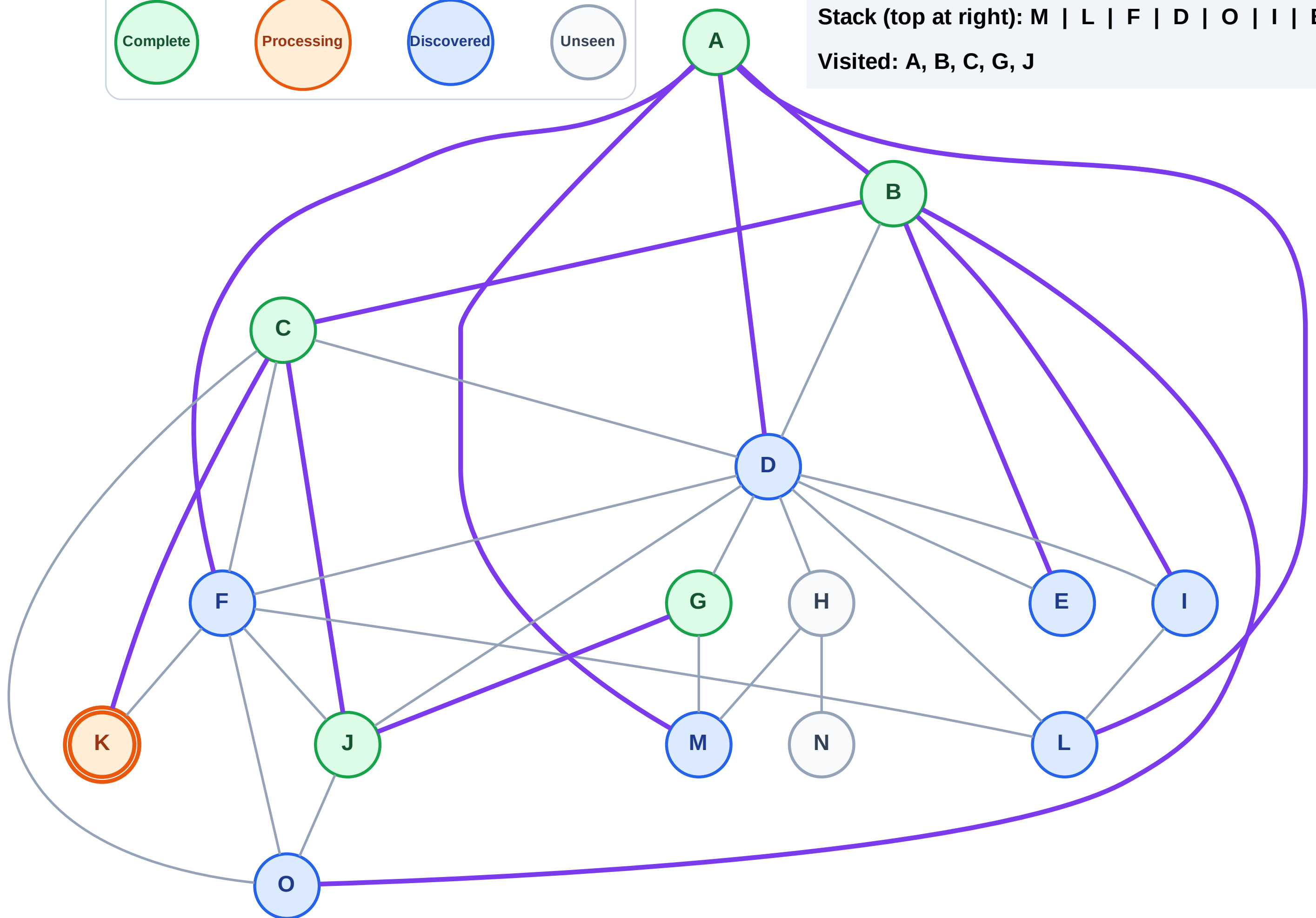
Discovered



Unseen

Stack (top at right): M | L | F | D | O | I | E

Visited: A, B, C, G, J



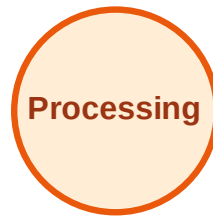
DFS Traversal

Step 13: TARGET NODE K REACHED! Path (3 hops): A → B → C → K

Legend



Complete



Processing



Discovered



Unseen

Stack (top at right): M | L | F | D | O | I | E

Visited: A, B, C, G, J, K

